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**Advanced Hemodynamic Modelling and Simulation of Flow  
Diverters in the Treatment of Intracranial Aneurysms**

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*Dedication*



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# Chapter 1

## Numerical Simulations of Hemodynamics

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**Abstract**

*In the following chapter, the incompressible, single-phase Navier-Stokes equations are introduced as the backbone of the hemodynamics simulations. The equations are derived from the conservation of mass and momentum and are further reformulated in a weak sense. The obtained formulation is then discretized with a Variational Multiscale-type Method (VMS) to stabilize the schemes in face of same-order test and trial functions for pressure and velocities. This additionally prevents spurious oscillations caused by convection dominated flows, such as the ones found in the large human arteries. The methodology is explained and validated on 2D and 3D test cases. The chapter is concluded with a discussion on the numerical results and the implications of the findings.*

**Resumé**

*Dans le chapitre suivant, les équations de Navier-Stokes incompressibles et monophasiques sont présentées comme l'épine dorsale des simulations hémodynamiques. Les équations sont dérivées de la conservation de la masse et de la quantité de mouvement et sont reformulées dans un sens faible. La formulation obtenue est ensuite discrétisée à l'aide d'une méthode variationnelle multiéchelle (VMS) afin de stabiliser les schémas face à des fonctions de test et d'essai de même ordre pour la pression et les vitesses. Cette méthode permet en outre d'éviter les oscillations parasites causées par les écoulements dominés par la convection, tels que les écoulements actuels. La méthodologie est expliquée et validée sur des cas d'essai en 2D et en 3D. Le chapitre se termine par une discussion sur les résultats numériques et les implications des conclusions.*

1

2 **1.1 The Incompressible Navier-Stokes Equations**

3 The Navier-Stokes equations stem, as mentioned before from the conservation of  
4 mass and momentum in an open and bounded domain  $\Omega \subset \mathbb{R}^n$  in which no material  
5 enters or leaves. For the former quantity, the conservation can be expressed through:

$$\frac{d}{dt} \int_{\Omega(t)} \rho dV = 0, \quad (1.1)$$

6 imposing that the integral of over the density  $\rho$  in the control volume does not  
7 change.

### 1.1.1 Reynolds Transport Theorem

The derivative over the time-dependent domain can be simplified thanks to the Reynolds transport theorem, named after its discoverer Osborne Reynolds.

**Theorem 1** (Reynolds Transport Theorem). *Let  $\vec{f}(t, \vec{x})$  be any conservation quantity, then the Reynolds transport theorem states that for a velocity field  $\vec{u}$ , the following holds:*

$$\frac{d}{dt} \int_{\Omega(t)} \vec{f} dV = \int_{\Omega(t)} \frac{\partial \vec{f}}{\partial t} + \nabla \cdot (\vec{u} \otimes \vec{f}) dV \quad (1.2)$$

*Proof.*

$$\frac{d}{dt} \int_{\Omega(t)} \vec{f}(t, \vec{x}) dV = \lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} \left( \int_{\Omega(t+\Delta t, \vec{x})} \vec{f}(t+\Delta t, \vec{x}) dV - \int_{\Omega(t)} \vec{f}(t, \vec{x}) dV \right) \quad (1.3)$$

To drag the derivative into the integral it is necessary to reformulate the problem in terms of a initial position  $\vec{X}$  and a mapping  $\vec{x} = \vec{\phi}(t, \vec{X})$ . With the corresponding derivative  $\nabla \vec{\phi} = F(t, \vec{X})$  and the Jacobian  $J = \det F$ , the integral can be rewritten in terms of the initial domain  $\Omega(0)$  and the contents can therefore be merged:

$$\frac{d}{dt} \int_{\Omega(t)} \vec{f} dV = \lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} \int_{\Omega(0)} \left[ (\vec{f} \circ \vec{\phi}) J \right] (t+\Delta t, \vec{X}) - \left[ (\vec{f} \circ \vec{\phi}) J \right] (t, \vec{X}) dV \quad (1.4)$$

$$= \int_{\Omega(0)} \left[ J \frac{\partial}{\partial t} (\vec{f} \circ \vec{\phi}) + (\vec{f} \circ \vec{\phi}) \frac{\partial J}{\partial t} \right] (t, \vec{X}) dV \quad (1.5)$$

To simplify the derivative of the Jacobian, recall Jacobi's formula.

$$\frac{\partial J}{\partial t} = J \operatorname{tr} \left( F^{-1} \frac{\partial F}{\partial t} \right) = J \operatorname{tr} \left( \frac{\partial}{\partial \vec{x}} \frac{\partial \vec{\phi}}{\partial t} \right) = J (\nabla \cdot \vec{u}) \quad (1.6)$$

Now the integral can be written back in terms of a time-dependent domain.

$$\frac{d}{dt} \int_{\Omega(t)} \vec{f} dV = \int_{\Omega(t)} \frac{\partial \vec{f}}{\partial t} + (\vec{u} \cdot \nabla) \vec{f} + \vec{f} (\nabla \cdot \vec{u}) dV = \int_{\Omega(t)} \frac{\partial \vec{f}}{\partial t} + \nabla \cdot (\vec{u} \otimes \vec{f}) dV \quad (1.7)$$

□

From now on all functions are taken to be time-dependent, as well as dependent on a location  $\vec{x}$  in an eulerian framework, unless stated otherwise. The explicit dependencies are omitted for the sake of brevity. Also, the derivatives in time will be denoted by  $\partial_t$ .

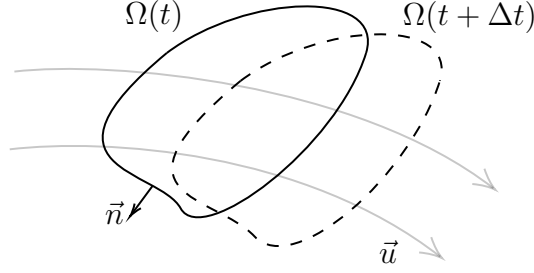


Figure 1.1: Illustration of the moving domain  $\Omega$ . In vague terms speaking, by sticking to the domain at time  $t$ , one needs to adjust the conservation laws to account for the displacement.

## 1.1.2 Conservation Equations

Using [Theorem 1](#), the conservation of mass can be expressed as:

$$\int_{\Omega} \partial_t \rho + \nabla \cdot (\rho \vec{u}) dV = 0 \quad (1.8)$$

The present integral form can be reduced to a differential form with a short proof by contradiction. Suppose that the content of the integral from [Eq. 1.8](#) is not zero in  $\Omega$ , then the integral would be adding positive and negative contributions over the domain. However, since the choice of  $\Omega$  is arbitrary, the integral formulation has to hold for any subdomain of  $\Omega$ , including those in which the integrand is non-zero. With the contradiction, the integrand has to be zero in  $\Omega$  and the differential form is obtained:

$$\partial_t \rho + \nabla \cdot (\rho \vec{u}) = 0 \quad (1.9)$$

In the same manner, the conservation of momentum is derived from preserving the volume-specific momentum  $\rho \vec{u}$ , which is modulated by body forces, such as the gravity, and stresses. The latter are the result of the interactions between the fluid particles and are expressed through the stress tensor  $\sigma$ . The conservation of momentum is then:

$$\partial_t (\rho \vec{u}) + \nabla \cdot (\rho \vec{u} \otimes \vec{u}) = \nabla \cdot \sigma + \rho \vec{g} \quad (1.10)$$

The conservative formulation shown in [Eq. 1.10](#) can be rewritten into the convective form by using the product rule and the conservation of mass.

$$\rho \partial_t \vec{u} + \rho (\vec{u} \cdot \nabla) \vec{u} = \nabla \cdot \sigma + \rho \vec{g} \quad (1.11)$$

The stress tensor  $\sigma$  is taken to be the:

$$\sigma = -pI + \lambda(\nabla \cdot \vec{u})I + 2\mu\varepsilon(\vec{u}) \quad (1.12)$$

43

$$\varepsilon(\vec{u}) = \frac{1}{2}(\nabla \vec{u} + \nabla \vec{u}^T) \quad (1.13)$$

|           |                   |      |
|-----------|-------------------|------|
| $p$       | Pressure          | Pa   |
| $\lambda$ | Lamé parameter    | Pa·s |
| $\mu$     | Dynamic viscosity | Pa·s |

Table 1.1: Physical quantities of stress tensor.

44 As will be more thoroughly discussed in the coming section, blood is incompressible  
45 at physiological pressures and temperatures, implying  $\rho = \text{const.}$  Therefore, one  
46 may proceed with the simplification of the momentum and mass equations:

$$\begin{aligned} \rho(\partial_t \vec{u} + (\vec{u} \cdot \nabla) \vec{u} - \vec{g}) &= -\nabla p + \nabla \cdot 2\mu \varepsilon(\vec{u}) \\ \nabla \cdot \vec{u} &= 0 \end{aligned} \quad (1.14)$$

### 47 1.1.3 Boundary Conditions

48 Boundary conditions have been identified by several publications and computing  
49 challenges as sensitive components of hemodynamics simulations [1, 2]. For that  
50 reason, Chapter 3 dedicates a section to inlet and outlet BCs, providing both  
51 measurement-based techniques and heuristics that depend on the vascular topol-  
52 ogy. In Chapter 4, more detail will be provided on the modelling of endoluminal  
53 devices. For now, they will be just shortly introduced in order to carry on with the  
54 derivation of the numerical framework.

55

56 The domain boundaries for segments of the brain arterial system are divided into  
57 four groups: inlets  $\Gamma_{in}$ , outlets  $\Gamma_{out}$ , walls  $\Gamma_w$  and endoluminal devices  $\Gamma_{fd}$ . In  
58 general terms, the inflow conditions, *i.e.* the blood supplied to the portion of the  
59 vasculature that is simulated, is imposed through a time-varying velocity profile. In  
60 mathematical terms speaking, it is expressed as:

$$\vec{u} = \vec{u}_{in}, \quad (\vec{x}, t) \in \Gamma_{in} \times (0, T] \quad (1.15)$$

61 Outflow conditions, on the other hand, are related to the flow split caused by the  
62 resistance and capacitance of the distal vasculature. The split can be either modelled  
63 as an imposed volumetric flow or a normal stress. For now, we will stick to the latter:

$$\sigma \cdot \vec{n} = \vec{h}, \quad (\vec{x}, t) \in \Gamma_{out} \times (0, T] \quad (1.16)$$

Arterial walls and endoluminal devices are modelled as rigid, no-slip boundaries. Therefore, in an Eulerian framework, the velocity at the vessel wall and the interface of any implanted device will be dictated by:

$$\vec{u} = 0, \quad (\vec{x}, t) \in \Gamma_w \cup \Gamma_{fd} \times (0, T] \quad (1.17)$$

The boundary conditions presented so far can be divided into two groups, essential on  $\Gamma_d$  and natural on  $\Gamma_n$ . These categories are relevant in Finite Element Methods (FEM), since their imposition leads to different formulations of the problem. Therefore, inflow, walls and any endoluminal devices are essential, while the outflow is natural.

The initial conditions can be set by previously solving a steady-state simulation at the inflow rate of interest. Other approaches, involve solving a potential flow or simply setting the initial solution to zero. For any of the choices, it is required to run simulations for several cardiac cycles to ensure that the initial transient is washed out and a periodic state has been reached.

$$\vec{u} = \vec{u}_0, \quad \vec{x} \in \Omega, t = 0 \quad (1.18)$$

## 1.2 Blood Rheology

Blood is a complex dense suspension comprising approximately 40–45% cellular components, including Red Blood Cells (RBCs), white blood cells (leukocytes), and platelets. The remainder of blood is called plasma, a homogeneous yellow substance composed of water (90% of plasma [3]), electrolytes, proteins, glucose, lipids, hormones, and carbon dioxide. Due to the high water content, both in cells and plasma, blood is generally considered incompressible, with a density of  $1056 \text{ kg/m}^3$  [4]. Likewise, plasma exhibits nearly Newtonian characteristics [5] with dynamic viscosities that only decrease from  $\mu_{\text{plasma}} = 1.27 \text{ mPa}\cdot\text{s}$  in a resting state to  $\mu_{\text{plasma}} = 1.23 \text{ mPa}\cdot\text{s}$  under high shearing [6].

RBCs account for more than 99% of the cells found in a blood stream and play the pivotal role of transporting oxygen in the body [3]. Their interior is filled with hemoglobin, an iron-rich protein with the capacity to store and release oxygen based on pressure levels and the chemistry of its environment. These cells are optimised for their transport task to the point in which they eject their nucleus to accommodate more space for hemoglobin and improve their flexibility. While resting, RBCs attain a biconcave disk shape ( $d = 8 \mu\text{m}$ ), as displayed in Fig. 1.2, but can change their shape elastically under stress in order 1 deformations. When blood flows, especially in areas of high shear stress, RBCs tend to orient themselves and elongate along the flow direction, allowing them to traverse narrow vessels [7]. This adaptability is particularly valuable in capillaries, whose diameter of  $5 \mu\text{m}$  lies below the diameter of RBCs [5].

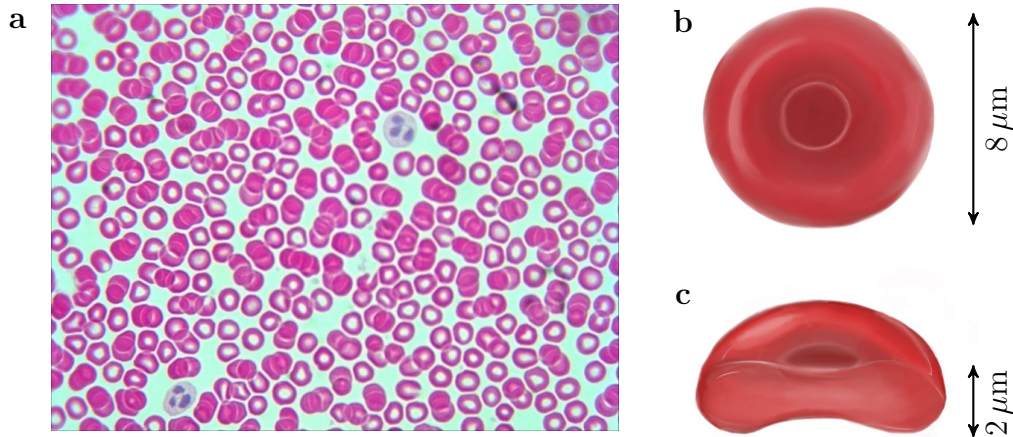


Figure 1.2: Red Blood Cells (RBCs). A microscope image of whole blood (a) from [8], the top view of a single RBC (b) and cross-sectional view (c).

### 1.2.1 Shear-thinning behaviour

Whole blood can display a complex flow behaviour due to the multiple and diverse components that it carries. At a macroscopic level, *i.e.* considering blood a single-phase medium, the apparent viscosity  $\mu$  grows in the presence of more RBCs, like in most suspensions. The concentration of RBCs is formally called hematocrit (Hct) and is given as a percentage of total blood volume. The Hct typically lays around 40–45%, depending on sex, but may rise or fall in an individual based on hydration levels and age factors. Chien *et al.* [9] confirmed the upward trend in viscosity for higher concentrations of RBCs experimentally, which we display in Fig. 1.3. The apparent viscosity was also shown to be clearly shear-thinning in the tested range, with changes of about one order of magnitude. The answer to what causes this behaviour is multifaceted and could, when thoroughly addressed, cover whole books. Here, we will limit ourselves to briefly discussing some of the key points that are relevant in the flow regime that is studied afterwards, the brain arterial flow.

The unique rheology of blood is primarily governed by the deformability and dynamics of RBCs, along with their reciprocal interactions and the vascular environment. They can deform and align under flow conditions, often aggregating into stacks known as “rouleaux” through reversible protein interactions. These rouleaux contribute significantly to blood’s rheological properties, imparting a non-zero apparent yield stress and contributing to shear-thinning viscosity and hysteresis. This behaviour classifies blood as an elasto-viscoplastic fluid, where the structural integrity of rouleaux depends on plasma protein fibrinogen, crucial for cell-cell interactions [10].

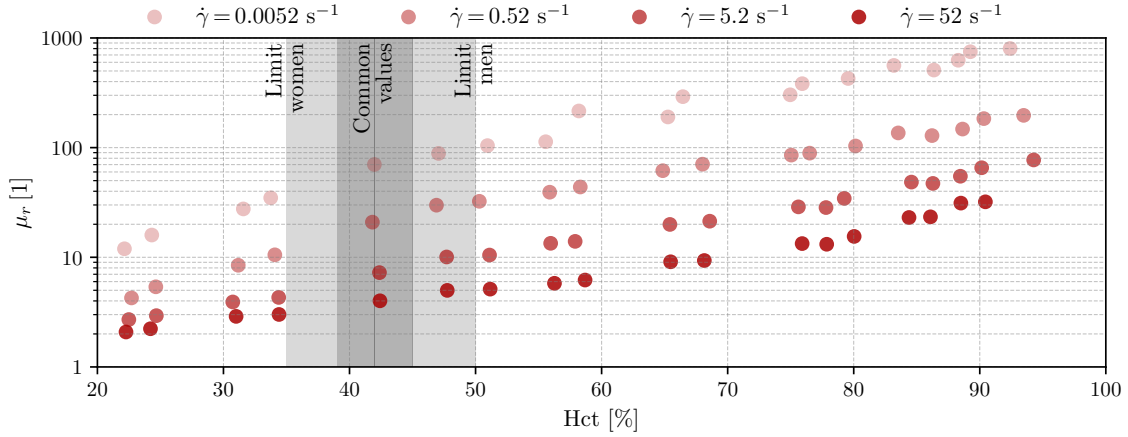


Figure 1.3: Relative apparent viscosity  $\mu_r = \mu/\mu_{\text{plasma}}$  of human RBCs suspended in blood plasma, as a function of their volume fraction, also called hematocrit (Hct), and shear rate [9]. The grey areas mark common Hct values for healthy adults. Women on average a concentration that is 4% lower than in man.

At low shear rates ( $\dot{\gamma} < 0.01 \text{ s}^{-1}$ ), numerous “rouleaux” stacks form in the suspension, leading to a significant increase in viscosity, up to 10 times higher compared to suspensions without aggregates [11]. In the intermediate shear rate range ( $0.01 < \dot{\gamma} < 10 \text{ s}^{-1}$ ), blood undergoes shear thinning due to the disintegration of the rouleaux stacks. Beyond a shear rate of approximately  $5\text{--}6 \text{ s}^{-1}$ , the viscosity curves of whole blood and RBC suspensions overlap, suggesting that viscosity is no longer primarily influenced by the aggregation of RBCs (see Fig. 1.4). Further shear-thinning occurs at higher shear rates ( $\dot{\gamma} > 100 \text{ s}^{-1}$ ), even after the rouleaux are disaggregated, as RBCs deform and align with the shear. This deformation contributes to rheofluidification in suspensions without aggregates and strongly depends on hematocrit, as shown by experiments in Fig. 1.3. Well above those higher shear rates ( $\dot{\gamma} > 1000 \text{ s}^{-1}$ ), the blood behaves as a Newtonian fluid.

In the human body shear rates vary strongly on the location and type of vessel. The largest shearing occurs typically in arterioles, where  $1600 \text{ s}^{-1}$  are commonly observed [12, 13]. Although this might sound counter-intuitive, the rough estimate using Poiseuille’s law reveals that the shear rate is inversely proportional to the cube of the vessel radius  $r$ ,  $\dot{\gamma} \propto 1/r^3$  in a volume drive flow, thus suggesting higher shearing in small caliber vessels. In this regime ( $d < 0.3 \text{ mm}$ ), blood displays the characteristic Fåhræus-Lindqvist effect [14] causing a drop in the apparent viscosity due to RBCs accumulating in the vessel centre and leaving a cell-free layer at the vessel wall. In this work, we do not consider this effect and restrict ourselves to vessels clearly above the critical  $0.3 \text{ mm}$  diameter mark. The largest artery in the

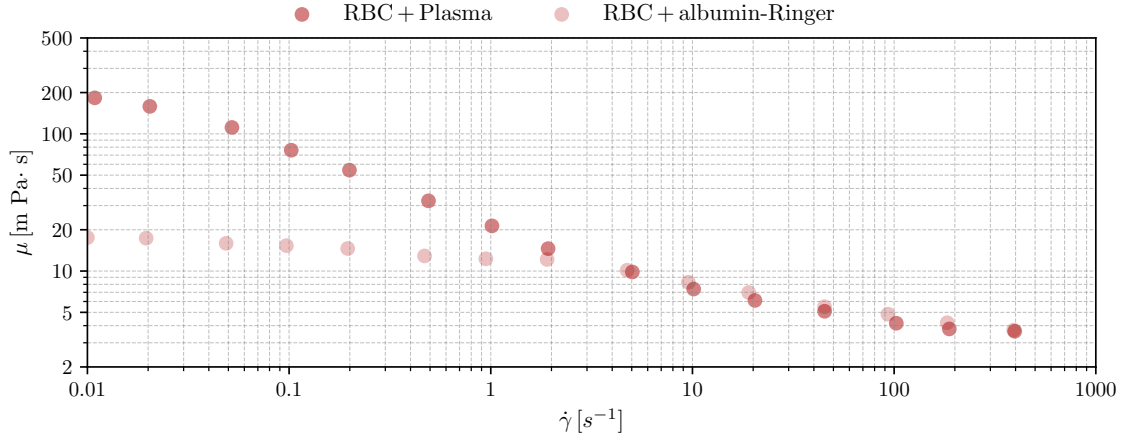


Figure 1.4: Measurement data from Chien et al. [11] for (heparin-treated) whole blood ( $Hct = 45\%$ ) and for RBCs in an 11% albumin-Ringer solution. This solution is commonly used to treat patients after or during a strong blood loss and reduces RBC aggregation, leading to a smaller apparent viscosity at low shearing.

body, the aorta features shear rates around  $150\text{ s}^{-1}$ , while its venous counterpart, the vena cava, only presents  $\dot{\gamma} = 9\text{ s}^{-1}$  due to its overall smaller flow velocity [12, 13]. In the case of brain arteries, Reynolds numbers and shear rates lay within the range spanned by arterioles and the aorta. Maximal values for shearing are usually observed at bifurcations, which are very common in the brain. When considering aneurysms however, the lower bound for  $\dot{\gamma}$  falls by an order of magnitude. In the cases of giant side-wall aneurysms ( $> 20\text{ mm}$ ), as will be shown farther below in Chapter 5, the flow can become notoriously slow leading to shear rates in the range of  $1\text{--}10\text{ s}^{-1}$ . In this regime assuming a constant viscosity is arguably problematic and is best avoided by considering a more complex rheology model.

## 1.2.2 Rheological Models

In practice, blood rheology in and around intracranial aneurysms can be modelled through a number of mathematical functions (Casson, Carreau-Yasuda, Power-Law, etc). The flexibility in the modelling comes from the lower bound of shear rates that we consider here, which, as a reminder, lays around  $1\text{ s}^{-1}$ . Above this threshold, many complex phenomena, such as elasticity of blood, are overruled by other flow properties allowing several equations to match a decaying profile. As a consequence, some model parameters, such as the relaxation time, may be greatly overestimated since they have no physical meaning with the data they are fit on. For a better display, Fig. 1.5 shows how the two models that we will discuss further below behave

over a spectrum of shear rates and, particularly, over the range of expected values for the simulations. We will also briefly discuss the otherwise omnipresent Newtonian assumption on IA simulations for completeness.

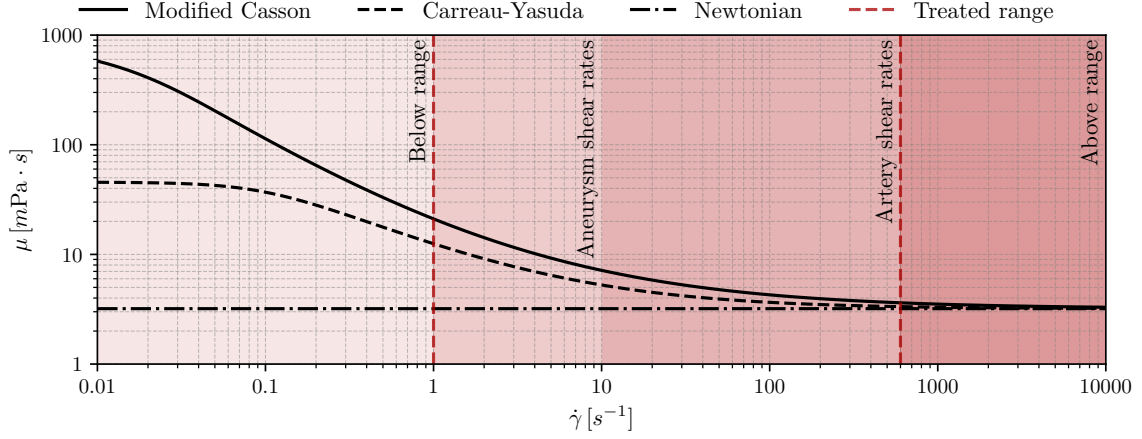


Figure 1.5: Comparison of Newtonian, Carreau-Yasuda, and Modified Casson models in the shear rate range of  $1 \text{ s}^{-1}$  to  $500 \text{ s}^{-1}$ . Lower shear rates ( $1$  to  $10 \text{ s}^{-1}$ ) are typical in aneurysms, while higher shear rates ( $10$  to  $500 \text{ s}^{-1}$ ) occur in arteries. The models differ significantly at lower shear rates, but converge at higher shear rates. This explains why there is a number of accepted models for single phase blood flow simulations.

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## 167 Newtonian Model

Newtonian behaviour is adopted by the overwhelming majority of simulation studies, both in large vessels and in aneurysms. Common values for the employed viscosity are between  $3.5$  and  $4 \text{ mPa}\cdot\text{s}$  [1, 2, 15]. In literature, several publications have addressed the question whether Newtonian assumptions are reasonable or not for computing the Wall Shear Stress (WSS) in IA with mixed conclusions. One part of the community [4, 16, 17] argues that prescribing a constant viscosity despite physical evidence against it leads to relevant hemodynamic changes, even in aortic flow. In aneurysm studies, changes were mostly of quantitative nature, with WSS being underestimated by 30% in slow flow regions. Khan *et al.* [18] challenged this idea and conducted an analysis including the comparison of first order solvers and their higher order solver *Oasis*. While confirming some of the trends in WSS from the other publications, they reported that the solver choice played a larger role in the WSS prediction. The concerns about the simulation fidelities are very valid, as well as the implications of boundary conditions on the hemodynamics. However, in the quest of tackling shear-thinning models, there seems to be a misinterpretation of

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the phenomena causing the drop of viscosity. The authors suggest that the aggregation of RBCs is the leading cause for the shear-thinning behaviour in aneurysm hemodynamics. However, as already described by Chien *et al.* [11], the aggregation of RBCs is a phenomenon that occurs at very low shear rates, typically below  $1 \text{ s}^{-1}$ . In the range of shear rates that are common for aneurysms, the viscosity of blood is mostly guided by the alignment of RBCs instead of aggregation. While transient phenomena of the viscosity are surely an important topic, this criticism could be unjustified. Nevertheless, Khan *et al.* [18] mention that the Newtonian assumption is probably not well suited for treated IA, where flow velocities are substantially smaller. The take from this discussion is that (i) numerical viscosities are important to be kept in check to avoid erroneous results and that (ii) no good argument for choosing a Newtonian rheology over a shear-thinning one is given. For this reason we will pursue the modelling of blood as a shear-thinning fluid.

## Carreau-Yasuda

The Carreau-Yasuda model is widely used to describe the non-Newtonian behaviour of blood. It captures the shear-thinning behaviour through the following equation:

$$\mu(\dot{\gamma}) = \mu_{\infty} + (\mu_0 - \mu_{\infty}) \left(1 + (\lambda \dot{\gamma})^2\right)^{(n-1)/2} \quad (1.19)$$

The parameter  $\mu_{\infty}$  denotes the viscosity at an infinite shear rate, which is the minimum viscosity the fluid can attain. Conversely,  $\mu_0$  is the viscosity at  $\dot{\gamma} = 0$ . The parameter  $\lambda$  is the relaxation time, which indicates how quickly the viscosity responds to changes in the shear rate. The power-law index  $n$  determines the degree of shear-thinning behaviour if  $n$  is less than 1, or shear-thickening behaviour if  $n$  is greater than 1. For  $n = 1$ , the model reduces to a Newtonian fluid behaviour, where viscosity remains constant regardless of  $\dot{\gamma}$ . For the specific case of blood, the constants provided are  $\mu_{\infty} = 0.0032 \text{ Pa}\cdot\text{s}$ ,  $\mu_0 = 0.0456 \text{ Pa}\cdot\text{s}$ ,  $\lambda = 10.03 \text{ s}$ , and  $n = 0.344$ . These values were determined through experimental measurements (Hct = 40%) and was obtained by non-linear least squares fit [19].

## Casson

An alternative to Carreau-Yasuda is the Casson model expressed by the following equation:

$$\mu = \frac{(\sqrt{\tau_y} + \sqrt{\dot{\gamma} \mu_C})^2}{\dot{\gamma}}$$

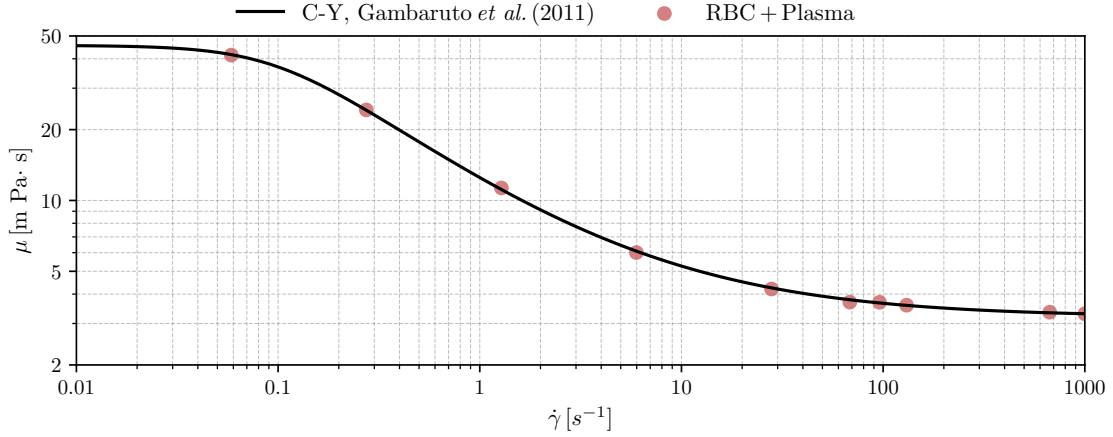


Figure 1.6: Fitted Carreau-Yasuda model with data from Gambaruto et al. [19]. The experimental data (in red) were obtained from a patient with an hematocrit of 40%.

Here,  $\tau_y$  represents the yield stress and  $\mu_C$  the Casson viscosity. This model provides a good fit for the viscoplastic behaviour of blood, where a certain threshold stress  $\tau_y$  must be exceeded before the fluid starts to flow. To address numerical challenges, such as division by zero at very low shear rates, a modified version of the Casson model has been introduced. The modified Casson model incorporates a new parameter,  $m$ , which acts as an apparent relaxation time, governing the transition of viscosity as the shear rate tends to zero:

$$\mu = \left( \sqrt{\mu_C} + \sqrt{\tau_y \frac{(1 - \exp(-m\dot{\gamma}))}{\dot{\gamma}}} \right)^2 \quad (1.20)$$

The parameter  $m$  ensures a smooth transition and avoids singularities in the model. A larger value of  $m$  implies a longer apparent relaxation time, leading to a slower decrease in viscosity with increasing shear rate, akin to a slower response of the material to applied shear stress.

Arguably, for our flow regime the Carreau-Yasuda model can replicate the Casson (see Fig. 1.5). The reason for still considering this new model is the well-documented fitting of its parameters to the Hct and their thorough validation in a recent study [4]. There samples of whole blood of 15 IA patients at 310K were analyzed and the dependencies shown in Eq. 1.21 were obtained. The Hct values found in the study range from 32% to 44%, and should therefore be taken with consideration for larger concentrations. In Fig. 1.7, the measurements and three sample curves at different Hct are plotted.

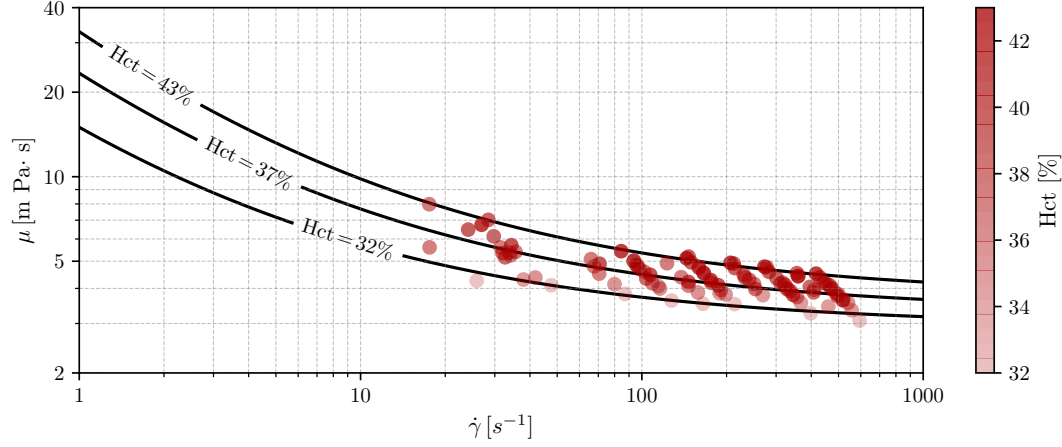


Figure 1.7: Experimental measurement by Suzuki et al. [4] for three example Hematocrit values (Hct). The colorbar on the right describes the hematocrit of the extracted samples, who belong to different patients with intracranial aneurysms.

$$\begin{aligned}\mu_C &= (0.073 \cdot \text{Hct} + 0.599) \text{ mPa} \cdot \text{s} \\ \tau_y &= (0.888 \cdot \text{Hct} - 23.753) \text{ mPa}\end{aligned}\tag{1.21}$$

### Conclusions

Blood rheology is a complex subject due to (i) the complexity of the material and (ii) the difficulties recording its properties. As discussed in experimental publications [4], even the anticoagulants used to prevent thrombosis in the extracted samples can have a noticeable impact on the viscosity of blood. Even though these effects perturb recording the viscoelastic properties of the suspension, its effects at least seem to be minor in the treated regime. For the further study, both the Carreau-Yasuda and the Casson model will be considered, with the latter being the preferred choice whenever Hct measurements are available.

231

## 1.3 Stabilized Finite Element Methods

232

To properly understand the content of the following section, it is important to revise some basics required in FEM for those who are not yet familiar with them. Contrary to finite differences and volumes, where the strong form of the equations can be used to start the discretization, finite elements base themselves on a weak formulation of the problem. This means that the strong form of the PDEs, like for instance

Eq. 1.14 is multiplied by a set of test functions from a suitable function space and integrated over the domain  $\Omega$ . By applying the chain rule of differentiation and the Gauß-divergence theorem, one can conveniently shift derivatives from the solution functions to the newly added test functions. The resulting integral formulations therefore require the solutions and test functions to be contained in a suitable function space, which is usually a Sobolev space. As a reminder, the Sobolev space  $H^1(\Omega)$  is the space of square integrable functions whose first derivatives are also square integrable. In more mathematical terms this can be written as:

$$H^1(\Omega) = \{v \in L^2(\Omega) \mid \nabla v \in L^2(\Omega)\}$$

$$L^2(\Omega) = \left\{v \mid \int_{\Omega} v^2 dV < \infty\right\}$$

Unless stated otherwise, the components of the continuous velocity  $\vec{u}$  and the test functions  $\vec{v}$  will be elements of  $H^1(\Omega)$ . With slight abuse of notation, we will write that vectorial variables will be part of a function space, meaning that each of the components is an element of that function space. The pressure  $p$  and the test functions linked to the mass conservation  $q$  will be elements of  $L^2(\Omega)$ . Note that when a fixed velocity is imposed on  $\vec{u}$  at the boundary, the test functions  $\vec{v}$  are forced to be zero on the same nodes. Therefore, the test function space is also sometimes referred to as  $H_0^1(\Omega)$ . In our particular case, we will use a subsets of  $H^1(\Omega)$  and  $L^2(\Omega)$ ,  $V_h$  and  $Q_h$  to approximate the solutions on a valid partition  $\mathcal{T}_h$  of  $\Omega$ . The subset consists of piecewise linear functions and therefore they are defined as follows:

$$V_h = \{v_h \in H^1(\Omega) \mid v_{h|K} \in \mathbb{P}_1, \forall K \in \mathcal{T}_h\}$$

$$Q_h = \{q_h \in L^2(\Omega) \mid q_{h|K} \in \mathbb{P}_1, \forall K \in \mathcal{T}_h\}$$

### 233 1.3.1 Stability Issues

234 One common question for all those starting to work with FEM discretizations of the  
 235 Navier-Stokes equations is why do most schemes include some sort of stabilization.  
 236 The reason is twofold involving (i) first the compatibilities between the interpolations  
 237 of velocity and pressures and secondly (ii) the effects of convection dominated flows.  
 238 These two problems are thoroughly discussed in many prominent publications, such  
 239 as [20, 21].

### 240 Stable Interpolations in mixed problems

241 We will start by providing an illustrative example of the first problem arising in the  
 242 particular case of a P1-P1 coupling. For showing possible problems arising from  
 243 using the same order interpolations for velocities and pressures, we will consider the  
 244 following simple example. Let us consider a 2D domain  $\Omega \subset \mathbb{R}^2$  with a square shape

245 and a uniform mesh of splitted rectangular elements (see Fig. 1.8). On that domain,  
 246 it is sought to solve the Stokes problem with homogeneous boundary conditions:

$$\begin{aligned} \nabla p - \nabla \cdot 2\mu\varepsilon(\vec{u}) &= \vec{f} \\ \nabla \cdot \vec{u} &= 0, \quad \vec{x} \in \Omega \end{aligned} \quad (1.22)$$

247 Since the Stokes problem admits any constant being added to the pressure, it is said  
 248 to be unique up to a constant. Therefore, to close the system an auxiliary condition  
 249 is added constraining the integral constant to fulfill that the volume integral of the  
 250 pressure is zero:

$$\int_{\Omega} p \, dV = 0 \quad (1.23)$$

251 The weak formulation of the problem is obtained by multiplying the equations with  
 252 discrete test functions  $\vec{v}_h$  and  $q_h$  and integrating over  $\Omega$ . The domain itself has a  
 253 valid partition  $\mathcal{T}_h$  into triangular elements  $K$  on which piecewise linear functions are  
 254 defined following the classical  $\mathbb{P}_1$ -Lagrange elements.

$$\begin{aligned} \int_{\Omega} \vec{v}_h \cdot \nabla p_h - \vec{v}_h \cdot [\nabla \cdot 2\mu\varepsilon(\vec{u}_h)] \, dV &= \int_{\Omega} \vec{f} \cdot \vec{v}_h \, dV, \quad \vec{v}_h \in V_h \\ \int_{\Omega} q_h \nabla \cdot \vec{u}_h \, dV &= 0, \quad q_h \in Q_h \end{aligned} \quad (1.24)$$

255 Using the Gauß-divergence theorem and the homogeneous boundary conditions, this  
 256 problem can be rewritten into:

$$\begin{aligned} \int_{\Omega} 2\mu\varepsilon(\vec{u}) : \varepsilon(\vec{v}) \, dV - \int_{\Omega} p \nabla \cdot \vec{v} \, dV &= \int_{\Omega} \vec{f} \cdot \vec{v} \, dV \\ \int_{\Omega} q \nabla \cdot \vec{u} \, dV &= 0 \end{aligned} \quad (1.25)$$

257 Assume now that  $\hat{p}_h$  as shown in Fig. 1.8 satisfies the relationship:

$$\int_{\Omega} \hat{p}_h \nabla \cdot \vec{v}_h \, dV = 0, \quad \forall \vec{v}_h \in V_h \quad (1.26)$$

Then it is clear that the solutions to the pressure of Eq. 1.25 are not unique, since any real-valued multiple of  $\hat{p}_h$  would also satisfy Eq. 1.26. In other words, the pressure is not uniquely determined by the velocity field, since for any  $\vec{v}_h$  the term vanishes. This can be easily verified by checking the veracity of  $\hat{p}_h$  as a possible solution. First,

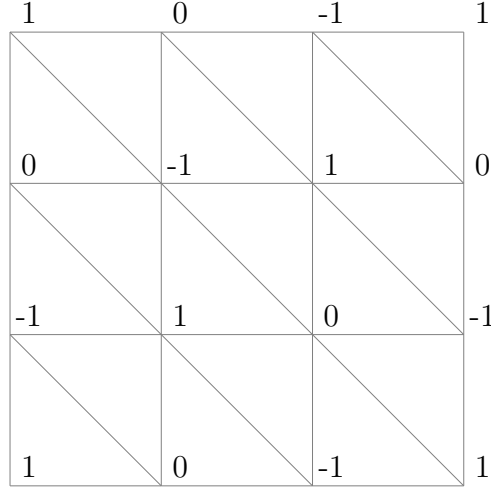


Figure 1.8: 2D unit square decomposed into triangular elements. A pressure solution is inscribed at the nodal positions of the mesh.

it is clear that  $\hat{p}_h \in Q_h$  and that the condition formulated in Eq. 1.23 is satisfied in all  $K \in \mathcal{T}_h$ .

$$\int_{\Omega} \hat{p}_h dV = \sum_{K \in \mathcal{T}_h} \int_K \hat{p}_h dV = \sum_{K \in \mathcal{T}_h} (-1 + 0 + 1) \int_K dV = 0$$

Then, it remains showing that the proposed  $\hat{p}_h$  is indeed in the kernel:

$$\begin{aligned} \int_{\Omega} \hat{p}_h \nabla \cdot \vec{v}_h dV &= \sum_{K \in \mathcal{T}_h} \int_K \hat{p}_h \nabla \cdot \vec{v}_h dV \\ &= \sum_{K \in \mathcal{T}_h} \nabla \cdot \vec{v}_{h,K} \int_K \hat{p}_h dV = 0, \quad \forall \vec{v}_h \in V_h \end{aligned}$$

258 Therefore, we can see that for the given mesh there is no uniqueness to the solutions,  
 259 posing a major problem. This phenomenon is due to the inherent compatibility  
 260 of equal order interpolations on some meshes, although it is not necessary, that  
 261 for all triangulations this pressure problem arises. This is more formally expressed  
 262 through the Inf-Sup condition, also called the Ladyzhenskaya-Babuska-Brezzi (LBB)  
 263 condition. The LBB condition is a necessary and sufficient condition for the stability  
 264 of the mixed finite element method for the Stokes and other problems that take a  
 265 similar shape. The LBB condition is met if there exist a mesh independent factor  
 266  $\beta > 0$  such that the following inequality holds:

$$\exists \beta > 0 : \quad \inf_{q_h \in Q_h} \sup_{\vec{v}_h \in V_h} \frac{\int_{\Omega} q_h \nabla \cdot \vec{v}_h dV}{\|q_h\|_{L^2} \|\vec{v}_h\|_{H^1}} \geq \beta > 0 \quad (1.27)$$

267 The condition (1.27) is satisfied unconditionally for some pairings, such as the Taylor-  
 268 Hood elements (P2-P1), but not for P1-P1 pairings as shown here. The stabilization  
 269 techniques introduced later however mitigate this inherent issue.

## 270 Convection dominated processes

271 Compared to the Stokes problem, the Navier-Stokes equations introduce the com-  
 272 plexity of a non-linear convection term  $(\vec{u} \cdot \nabla)\vec{u}$ . In physical terms speaking it  
 273 describes how certain information is carried by a velocity field throughout the do-  
 274 main. To understand the challenges it involves it suffices again to consider a much  
 275 simpler example. Take the stationary advection-diffusion equation in 1D for any  
 276 scalar field  $\varphi$  and a positive diffusion coefficient  $\varepsilon$ . Let  $u$  be a constant velocity.

$$\begin{aligned} u\partial_x\varphi - \varepsilon\partial_{xx}\varphi &= f \\ \varphi(\Gamma_0) &= \varphi(\Gamma_1) = 0 \end{aligned} \quad (1.28)$$

277 One can interpret  $\varphi$  as to be a temperature that is convected through a pipe to  
 278 transport heat between sources and sinks given by  $f$ . The boundaries are set to the  
 279 reference temperature 0. Here again we compute the weak formulation:

$$\int_{\Omega} u\psi\partial_x\varphi + \varepsilon\partial_x\varphi\partial_x\psi \, dx = \int_{\Omega} f\psi \, dx, \quad \forall \psi \in H_0^1(\Omega) \quad (1.29)$$

280 The bilinear operator  $a(\cdot, \cdot)$  of Eq. 1.29 can be written as:

$$a(\varphi, \psi) = \int_{\Omega} u\psi\partial_x\varphi + \varepsilon\partial_x\varphi\partial_x\psi \, dx \quad (1.30)$$

281 With this bilinear form, we are going to show that for the standart Galerkin dis-  
 282 cretization some necessary criteria for existence and uniqueness are not met when  
 283  $\varepsilon \rightarrow 0$ . Before continuing, the Lax-Milgram theorem is stated:

**Theorem 2** (Lax-Milgram). *Let  $V$  be a Hilbert space and  $a(\cdot, \cdot)$  a bilinear form on  $V$  that satisfies:*

$$\text{Continuity: } \exists C \in \mathbb{R}^+ : |a(u, v)| \leq C \|u\|_V \|v\|_V, \quad \forall u, v \in V$$

$$\text{Coercivity: } \exists \alpha \in \mathbb{R}^+ : a(u, u) \geq \alpha \|u\|_V^2, \quad \forall u \in V$$

*Then there exists a unique solution  $u \in V$  for a bounded linear form  $f(v)$  to the variational problem:*

$$a(u, v) = f(v), \quad \forall v \in V$$

284

In the convection-diffusion equation, the coercivity condition reads as followed:

$$a(\varphi, \varphi) = \int_{\Omega} u \varphi \partial_x \varphi + \varepsilon (\partial_x \varphi)^2 dx \quad (1.31)$$

$$= \int_{\Omega} \frac{u}{2} \partial_x \varphi^2 + \varepsilon (\partial_x \varphi)^2 dx \quad (1.32)$$

$$= \frac{u}{2} [\varphi^2]_{\Gamma_0}^{\Gamma_1} + \int_{\Omega} \varepsilon (\partial_x \varphi)^2 dx \quad (1.33)$$

$$= \varepsilon \|\varphi\|_{H^1(\Omega)}^2 \quad (1.34)$$

285 Therefore, for  $\varepsilon \rightarrow 0$  or in other words, for pure convection the coercivity condition  
 286 is not met and uniqueness and existence cannot be guaranteed. This phenomenon  
 287 can also be observed in finite differences when central schemes are used for the  
 288 space discretization of convective terms. In fact, a discretization with central finite  
 289 differences of the convection-diffusion problem will lead to the same linear system  
 290 with first order Lagrange elements. For that reason, the Gakerkin FEM is oftentimes  
 291 seen as a central scheme. In finite difference discretizations, the problem is usually  
 292 tackled by exchanging higher order central schemes by upwind schemes (UD). These,  
 293 as the name suggests, employ the data found upwind of the flow velocity to discretize  
 294 spatial gradients.

$$\partial_x^{UD} \varphi = \begin{cases} \frac{\varphi(x) - \varphi(x - \Delta x)}{\Delta x}, & u > 0 \\ \frac{\varphi(x + \Delta x) - \varphi(x)}{\Delta x}, & \text{else} \end{cases} + \mathcal{O}(\Delta x) \quad (1.35)$$

An interesting note is, that these schemes can also be interpreted as central differences with added diffusion terms. Consider for example the case  $u > 0$ :

$$\begin{aligned} \partial_x \varphi &= \frac{\varphi(x + \Delta x) - \varphi(x - \Delta x)}{2\Delta x} + \mathcal{O}(\Delta x^2) \\ &\approx \frac{\varphi(x) - \varphi(x - \Delta x)}{\Delta x} + \frac{\Delta x}{2} \left( \frac{\varphi(x + \Delta x) - 2\varphi(x) + \varphi(x - \Delta x)}{\Delta x^2} \right) \\ &= \frac{\varphi(x) - \varphi(x - \Delta x)}{\Delta x} + \mathcal{O}(\Delta x) \\ \partial_x^{UD} \varphi &= \partial_x \varphi - \varepsilon \partial_{xx} \varphi, \quad \varepsilon = \frac{\Delta x}{2} \end{aligned}$$

295 Thus, a first order upwinding scheme can be seen as a central scheme with an added  
 296 diffusion term of the order  $\varepsilon$  to satisfy the coercivity constraint. Luckily, the VMS-  
 297 type stabilization that we will employ later does also tackle this problem. While the  
 298 compatibility constraint is alleviated through a relaxation of the mass balance, the  
 299 convection phenomena are handled by a resulting shift in the test function. This  
 300 shift, as will be seen later, is proportional to the convective velocity and acts in a  
 301 similar way to a flux delimiter.

### 1.3.2 Variational Multiscale Method

One strategy that allows for a stable approximation of the solution of the Navier-Stokes equations for same order Lagrange elements under the presence of large convection terms is the Variational MultiScale method (VMS). VMS-type discretizations have in common that they split the solution into a resolved large scale  $\vec{U}_h$  and a subgrid contribution  $\vec{U}'$ . While the large scales are solved, the subgrid scales are usually modelled, incorporating a stabilizing effect into the system [22, 23]. This idea shares many similarities with other turbulence models, such as the Large Eddy Simulations (LES) or the Reynolds Averaged Navier-Stokes (RANS). In fact, it can be shown, that the stabilization technique that is going to be introduced is closely related to an LES approach [23]. As hinted before, there are many VMS formulations for the Navier-Stokes equations, each of which stems from different assumptions made during the derivation. To put into context the VMS-type stabilization that is used here, we will derive the method from the most general point of view and discuss the key assumptions at the moment when they are introduced. For this purpose, the variables  $\vec{u}$  and  $p$  are bundled into a single vector  $\vec{U} = [\vec{u}, p]^T$ , as well as the respective test functions of the momentum and mass equations  $\vec{V} = [\vec{w}, q]^T$ . We also introduce also the Navier-Stokes operator  $\mathcal{L}$  and the right-hand side  $\vec{F}$  in a notation consistent with [22].

$$\mathcal{L}(\vec{u}_{conv}, \vec{U}) = \begin{bmatrix} \rho \partial_t \vec{u} + \rho(\vec{u}_{conv} \cdot \nabla) \vec{u} + \nabla p - \nabla \cdot 2\mu \varepsilon(\vec{u}) \\ \nabla \cdot \vec{u} \end{bmatrix}, \quad \vec{F} = \begin{bmatrix} \rho \vec{g} \\ 0 \end{bmatrix} \quad (1.36)$$

Therefore, the strong form of the Navier-Stokes Equations can be equally written as:

$$\begin{aligned} \mathcal{L}(\vec{u}_{conv}, \vec{U}) &= \vec{F}, & (\vec{x}, t) &\in \Omega \times (0, T] \\ \vec{u} &= \vec{u}_0, & \vec{x} &\in \Omega, t = 0 \\ \sigma \cdot \vec{n} &= \vec{t}_n, & (\vec{x}, t) &\in \Gamma_n \times (0, T] \\ \vec{u} &= \vec{u}_d, & (\vec{x}, t) &\in \Gamma_d \times (0, T] \end{aligned} \quad (1.37)$$

The weak form of the Navier-Stokes equations can be obtained by multiplying the strong form Eq. 1.37 with tests functions  $\vec{V}$  contained in the Sobolev space  $H^1(\Omega)$  and integrating the terms over  $\Omega$ .

$$\int_{\Omega} \vec{V} \cdot \mathcal{L}(\vec{u}_{conv}, \vec{U}) dV = \int_{\Omega} \vec{V} \cdot \vec{F} dV, \quad \forall \vec{V} \in H^1(\Omega) \quad (1.38)$$

VMS-type methods, as described above, now introduce a decomposition of the solution into a resolved part  $\vec{U}_h$  and a subgrid part  $\vec{U}'$ . The coarse scales are solved

on the existing mesh with, in our case, linear elements, while the subgrid scales are the infinite-dimensional and will be modelled instead. The same decomposition is applied to the test functions  $\vec{V}$ .

$$\begin{aligned}\vec{U} &= \vec{U}_h + \vec{U}', & \vec{U} \in W &=: \{\vec{u}, p \in H^1(\Omega) \mid \vec{u} = \vec{u}_d \text{ on } \Gamma_d\} \\ \vec{V} &= \vec{V}_h + \vec{V}', & \vec{V} \in W_0 &=: \{\vec{w}, q \in H^1(\Omega) \mid \vec{w} = \vec{0} \text{ on } \Gamma_d\}\end{aligned}\quad (1.39)$$

The functions  $\vec{U}_h$  and  $\vec{V}_h$  belong to the finite dimensional subspaces  $W_h$  and  $W_{h,0}$  of  $W$  and  $W_0$  that are described by classic linear elements on a triangulated mesh. The spaces of the subscales  $W'$ ,  $W'_0$  are therefore infinitely dimensional and complete the large scale space in  $W$  and  $W_0$ .

$$W' = W \setminus W_h, \quad W'_0 = W_0 \setminus W_{h,0} \quad (1.40)$$

If now the splitting is introduced into the weak form of the Navier-Stokes equations, the following equation is obtained:

$$\int_{\Omega} (\vec{V}_h + \vec{V}') \cdot \mathcal{L}(\vec{u}_{conv}, \vec{U}_h + \vec{U}') dV = \int_{\Omega} (\vec{V}_h + \vec{V}') \cdot \vec{F} dV \quad (1.41)$$

**The convective velocity**  $\vec{u}_{conv}$  was silently introduced before with the operator  $\mathcal{L}$  for a practical reason. In the most general form of the stabilizations,  $\vec{u}_{conv}$  is nothing but the velocity itself and thus when introducing the decomposition, the convective term produces cross terms. These cross terms are complicated to handle when computing an approximation for the subscales, since they are non-linear. Bazilevs *et al.* [20] studied the hypothetical case of using a perturbation series expansion to handle the non-linearity, coming to the conclusion that if the subscales are to be solved through Green's functions, only the first term of the series can be kept, *i.e.* the large scale velocity  $\vec{u}_h$ . Nevertheless, there are many publications that, while neglecting  $\vec{u}'$  in the subscale equations, keep it in the large scale equation yielding cross terms related to the ones from LES models. Since, to the writers knowledge, there is no general proof about the fact that the latter option bear any advantage, we will assume that the convective velocity is solely dictated by the large scales. Further, the convective velocity  $\vec{u}_h$  will be either determined by an iterative process, such as Newtons method, or treated semi-implicitly in transient simulations, allowing us to see the problem as a linear one for the derivation  $\mathcal{L}(\vec{u}_h, \cdot) \rightarrow \mathcal{L}(\cdot)$ . Since the weak formulation was posed for any test function, it has to hold for large and small scales independently. Therefore, we split the problem into:

$$\begin{aligned}\int_{\Omega} \vec{V}_h \cdot \mathcal{L}(\vec{U}_h + \vec{U}') dV &= \int_{\Omega} \vec{V}_h \cdot \vec{F} dV \\ \int_{\Omega} \vec{V}' \cdot [\mathcal{L}(\vec{U}_h + \vec{U}') - \vec{F}] dV &= 0\end{aligned}\quad (1.42)$$

355 **The orthogonality of the subscales** is now exploited to find an approximation  
356 of  $\vec{U}'$ . It can be seen that the subscale equation is satisfied if the terms multiplied by  
357 the subscale test function  $\vec{V}'$  result in an orthogonal function to  $W'_0$  in each partition  
358 of  $\Omega$  (Eq. 1.43).

$$\mathcal{L}(\vec{U}_h + \vec{U}') - \vec{F} = \vec{V}^\perp, \quad \vec{V}^\perp \in W_0^\perp \quad (1.43)$$

359 Codina *et al.* [22, 24, 25] studied  $\vec{V}^\perp$  as an  $L_2$  projection of the large scale resid-  
360 ual under different assumptions, generating so called Orthogonal Subscales. The  
361 additional term appearing in the subscales needs to be computed either implicitly  
362 by extending the existing linear system with an additional  $L_2$  projection [25] or ex-  
363 plicitly [22]. While the additional computational effort has been reported to be  
364 moderate, so seem its benefits [24, 25]. The diffusion induced by the stabilization  
365 is reduced, increasing accuracy on one side, but limiting stability on the other. For  
366 that reason, most VSM-type methods [20, 21, 23, 26] resort to the second strategy,  
367 namely assuming  $\vec{V}^\perp = 0$ .

$$\mathcal{L}(\vec{U}_h + \vec{U}') - \vec{F} = 0 \quad (1.44)$$

368 Eq. 1.44 is a special case of Eq. 1.43, called Algebraic Subgrid Scales (ASGS), in  
369 which the computation of the subgrid scales is simpler. Eq. 1.44 can be rearranged  
370 to separate the scales and introduce the residual  $\mathcal{R}$ .

$$\mathcal{L}(\vec{U}') = \vec{F} - \mathcal{L}(\vec{U}_h) = \mathcal{R}(\vec{U}_h) \quad (1.45)$$

371 Here it can be nicely seen, that the subscales represent in some form the errors found  
372 in the large scale solutions of the system. Conversely, finding an exact solution to  
373 the Navier-Stokes equations causes the subscales to vanish, which is a beneficial  
374 property of this approach.

375 **The temporal discretization of the subscales** is likewise a matter of debate  
376 that is best discussed at this point. Since the split of the solution into both scales  
377 introduces a time derivative of  $\vec{U}'$ , it is expected that an additional set of equations  
378 need to be integrated in time. In most VMS-type methods this term is neglected  
379  $\partial_t \vec{U}' \approx 0$  under the assumption of quasi-static subscales (QSS) [20, 26, 27]. It is  
380 argued, that the subscales react to the effects of the large scales and do not greatly  
381 change their behaviour due to their past. Others, have studied the effect of consid-  
382 ering the temporal derivatives, giving rise to the dynamics subscales [22, 28]. These  
383 have been shown to be more robust for different choices of timesteps and less prone  
384 to oscillations than the QSS counterpart [29]. Nevertheless, for the present work we  
385 will stick to the conventional treatment of the subscales as is mostly done in the  
386 hemodynamics field.

$$\vec{U}' = \mathcal{L}_{QSS}^{-1}[\mathcal{R}(\vec{U}_h)] \quad (1.46)$$

Expression (1.46) is the basis for the subscale modelling. The principal idea, is finding an appropriate description of the inverse operator  $\mathcal{L}_{QSS}^{-1}$ , so that  $\vec{U}'$  can be computed and re-injected into the large scale equation, which we have so far, not commented. Since the subscale modelling is a complex topic, an own subsection is dedicated to it.

### 1.3.3 Subscale Modelling

When deriving an expression for the subscales, we restrict the domain of each approximation to an element  $K \in \mathcal{T}_h$ . There are many different approaches to derive the same set of stabilization quantities, *e.g.* using Green's functions, Bubble elements and Fourier Transformations. The latter will be used for the demonstration following the main steps from [30] and [22]. On the subscales, we will assume viscosity variations to be negligible, *i.e.*  $\nabla \mu \approx 0$ . Additionally, all fluxes across the boundaries of  $K$  will be neglected on the subscales. We denote the residuals of the momentum and continuity equations as  $\vec{\mathcal{R}}_M$  and  $\mathcal{R}_C$ , respectively. Then, the problem to solve here is more precisely given by:

$$\begin{bmatrix} \rho(\vec{u}_h \cdot \nabla) \vec{u}' + \nabla p' - \mu \Delta \vec{u}' \\ \nabla \cdot \vec{u}' \end{bmatrix} = \begin{bmatrix} \vec{\mathcal{R}}_M \\ \mathcal{R}_C \end{bmatrix}, \quad \text{in } K \quad (1.47)$$

$$\vec{u}' = 0, \quad p' = 0, \quad \text{on } \partial K$$

As mentioned above, the derivation will use Fourier transformations of the unknown subscales, denoted by  $\mathcal{F}$  (here  $i$  represents  $\sqrt{-1}$ ).

$$\mathcal{F}\{g(\vec{x})\} = \int_K \exp(-i\vec{k} \cdot \vec{x}) g(\vec{x}) dV \quad (1.48)$$

The Fourier Transform of the partial derivatives are then computed with the following steps:

$$\begin{aligned} \mathcal{F}\left\{\frac{\partial g}{\partial x_j}\right\} &= \int_K \exp(-i\vec{k} \cdot \vec{x}) \frac{\partial g}{\partial x_j}(\vec{x}) dV \\ &= \oint_{\partial K} \exp(-i\vec{k} \cdot \vec{x}) g(\vec{x}) dA - \int_K -ik_j \exp(-i\vec{k} \cdot \vec{x}) g(\vec{x}) dV \\ &\approx ik_j \mathcal{F}\{g(\vec{x})\} \end{aligned} \quad (1.49)$$

Where we applied partial differentiation to obtain the intermediate step. The approximation comes from the assumed boundary conditions of problem (1.47). Also,

as pointed out in [22], it is expected that the subscales are characterized by large wave numbers  $\vec{k}$ , since lower frequencies are solved by the large scales. This gives dominance to the second term in the derivative of the Fourier Transform, limiting the action of boundary contributions. With this, we can proceed by reformulating problem (1.47) in terms of Fourier transformations:

$$\begin{aligned} \left[ \rho(\vec{u}_h \cdot i\vec{k}) + \mu(\vec{k} \cdot \vec{k}) \right] \hat{\vec{u}} + i\vec{k}\hat{p} &= \hat{\vec{\mathcal{R}}}_M \\ i\vec{k} \cdot \hat{\vec{u}} &= \hat{\mathcal{R}}_C \end{aligned} \quad (1.50)$$

By performing the same operation to the divergence of the momentum equation, we obtain an analogous of the Pressure Poisson Equation. It is now assumed without loss of generality, that the potential of the momentum residual is contained within the pressure subscale, allowing  $\nabla \cdot \mathcal{R}_M = 0$  [22].

$$\left[ \rho(\vec{u}_h \cdot i\vec{k}) + \mu(\vec{k} \cdot \vec{k}) \right] \hat{\mathcal{R}}_C - (\vec{k} \cdot \vec{k})\hat{p} = 0 \quad (1.51)$$

Here we also included the fact that  $i\vec{k} \cdot \hat{\vec{u}} = \hat{\mathcal{R}}_C$ . Then it is clear that the pressure subscale is approximated by:

$$\hat{p} = \left[ i\rho \frac{(\vec{u}_h \cdot \vec{k})}{(\vec{k} \cdot \vec{k})} + \mu \right] \hat{\mathcal{R}}_C \quad (1.52)$$

Contrary to what one might expect without previously knowing the derivation of the subscale model, the approximation for  $\hat{p}$  is not simply reinjected into the momentum equation [22]. This is mostly argued due to simplicity, as keeping the continuity residual  $\mathcal{R}_C$  in the velocity subscales would result in a non-SPD stabilization tensor (since there is no contribution of  $\vec{\mathcal{R}}_M$  on the pressure stabilization either). Instead the velocity subscales are given by:

$$\hat{\vec{u}} = \left[ i\rho(\vec{u}_h \cdot \vec{k}) + \mu(\vec{k} \cdot \vec{k}) \right]^{-1} \hat{\vec{\mathcal{R}}}_M \quad (1.53)$$

In [22], it is shown that by neglecting the boundary contributions (as done above) and applying consequently Plancherel's Theorem as well as the Mean Value Theorem, the stabilization parameters might be identified by the absolute values of the factors in front of the residuals and a mean value frequency ( $\vec{k}_0$ ):

$$\begin{aligned} \vec{u}' &\approx \left[ \rho^2(\vec{u}_h \cdot \vec{k}_0)^2 + \mu^2(\vec{k}_0 \cdot \vec{k}_0)^2 \right]^{-1/2} \vec{\mathcal{R}}_M := \tau_M \vec{\mathcal{R}}_M \\ p' &\approx (\vec{k}_0 \cdot \vec{k}_0)^{-1} \left[ \rho^2(\vec{u}_h \cdot \vec{k}_0)^2 + \mu^2(\vec{k}_0 \cdot \vec{k}_0)^2 \right]^{1/2} \mathcal{R}_C := \tau_C \mathcal{R}_C \end{aligned} \quad (1.54)$$

Through this derivation, the subscales have been modelled as piece-wise constant functions that act on each element individually. Therefore, they can be interpreted as bubble functions enriching the space of the large scales with their modelled effects.

### Reference Element Stabilization Tensor

The mean value frequency is an unknown, yet existing frequency for which the subscales are ideally solved. It is intuitive to think that it is related to some quantity proportional a derivative operator. These are proportional to the inverse meshsize along the direction of the gradient. In [22],  $\vec{k}_0$  is given as a product of the inverse mesh size measured by the perimeter of the element and a mesh independent factor. In other studies [30], the derivation is done right away from a reference domain, in this case the unit-reference element, obtaining an expression of the shape:

$$\tau_M = [\rho^2 \vec{u}_h^T G \vec{u}_h + \mu^2 G : G]^{-1/2} \quad (1.55)$$

Where  $G$  is the “metric tensor related to the isoparametric mapping” [30]. The velocity  $\vec{u}$  is interpolated to the cell centers from the nodes, where it resides. The illustrative derivation of [30] tackled the problem of subscales for the Convection Diffusion Reaction equations. In other works [20, 27], these are used in an extended form for the Navier-Stokes Equations as:

$$\begin{aligned} \tau_M &= \left[ \frac{4\rho^2}{\Delta t^2} + \rho^2 \vec{u}_h^T G \vec{u}_h + \mu^2 G : G \right]^{-1/2} \\ \tau_C &= \frac{1}{\tau_M \operatorname{tr} G} \end{aligned} \quad (1.56)$$

Consistent with our derivation, this method expresses the stabilization parameter  $\tau_C$  as a function of  $\tau_M$ . The timestep component now found in  $\tau_M$  is however an addition that in strict terms speaking should not appear in QSS. Since time component do not directly play a role following that assumption, they should neither appear in the computed approximation of the inverse operator. Unfortunately, both in tests ran for this work as well as in related literature [31, 31], not including the term in the stabilization leads in most engineering applications to convergence issues. Apart from this, we observe that a dominant fraction of the literature related to VMS for the transient Navier-Stokes Problem considers it [20, 27, 32, 33]. Naturally, including this member into the stabilization comes at a cost. Since the timestep is included with a inverse quadratic exponential, it will dominate the stabilization parameter for, *i.e.* for  $\Delta t \ll 1$ ,  $\tau_M \rightarrow \Delta t / \rho$ . Thus, the timestep becomes with this formulation a constraint in two ways: (i) large timesteps may not be sufficient to resolve certain phenomena in the flow and (ii) a too small timestep may force the stabilization to an insufficient level.

Returning to the proposed stabilization by [20], we likewise find an interpretation of the frequency vector  $\vec{k}$  as a mapping from the actual element, denoted by  $K$  shape to a reference element  $K_{ref}$ . The reference element is taken, in the context of  $P1$

463 elements, as to be the unit-triangle (see Fig. 1.9). More precisely and following the  
 464 description of [20], let one element  $K \in \Omega$  be determined by a set of nodes  $\vec{x}(i)$   
 465 for which a continuously differentiable bijective mapping  $\vec{x} = \vec{x}(\vec{\xi}) : K_{ref} \rightarrow K$  is  
 defined. Then the tensor  $G$  is defined by Eq. 1.57, that by construction is Symmetric

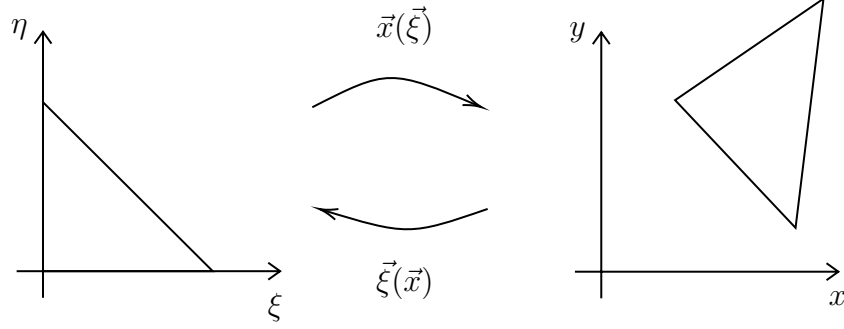


Figure 1.9: Transformation from unit-triangle (left) to real triangle  $K$  (right).

466 Positive Definite (SPD) and composed by constant entries for linear elements.  
 467

$$G = \frac{\partial \vec{\xi}}{\partial \vec{x}}^T \otimes \frac{\partial \vec{\xi}}{\partial \vec{x}} \quad (1.57)$$

In a series of implementations with the unit-triangle as a reference, we have encountered some difficulties that have lead to a more in depth analysis of the problem. At all points, we keep in mind that the definition of the methodology might have been misunderstood, regardless of the large number of publications giving similar descriptions on how to implement it. To show the key issues, take a simple 2D example:

$$\vec{x} = \vec{x}^1 + (\vec{x}^2 - \vec{x}^1)\xi + (\vec{x}^3 - \vec{x}^1)\eta \quad (1.58)$$

$$\frac{\partial \vec{x}}{\partial \vec{\xi}} = \begin{pmatrix} x^2 - x^1 & y^2 - y^1 \\ x^3 - x^1 & y^3 - y^1 \end{pmatrix} \quad (1.59)$$

$$\frac{\partial \vec{\xi}}{\partial \vec{x}} = \frac{\partial \vec{x}^{-1}}{\partial \vec{\xi}} = \left| \frac{\partial \vec{x}}{\partial \vec{\xi}} \right|^{-1} \begin{pmatrix} y^3 - y^1 & y^1 - y^2 \\ x^1 - x^3 & x^2 - x^1 \end{pmatrix} \quad (1.60)$$

468 Take a unit triangle stretched by a mesh size  $h$  as to be the element  $K$ .

$$x^1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad x^2 = \begin{pmatrix} h \\ 0 \end{pmatrix}, \quad x^3 = \begin{pmatrix} 0 \\ h \end{pmatrix} \quad (1.61)$$

469 For these points, the previously written transformations are:

$$\begin{aligned} \frac{\partial \vec{x}}{\partial \vec{\xi}} &= \begin{pmatrix} h & 0 \\ 0 & h \end{pmatrix}, & \frac{\partial \vec{\xi}}{\partial \vec{x}} &= \frac{\partial \vec{x}^{-1}}{\partial \vec{\xi}} = \begin{pmatrix} h^{-1} & 0 \\ 0 & h^{-1} \end{pmatrix} \\ G &= \begin{pmatrix} h^{-2} & 0 \\ 0 & h^{-2} \end{pmatrix}, & G : G &= 2h^{-4} \end{aligned} \quad (1.62)$$

470 Suppose now that instead of taking the point enumeration as before, the indices are  
471 permuted once, namely:

$$x^1 = \begin{pmatrix} h \\ 0 \end{pmatrix}, \quad x^2 = \begin{pmatrix} 0 \\ h \end{pmatrix}, \quad x^3 = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (1.63)$$

472 Then the metrics change to:

$$\begin{aligned} \frac{\partial \vec{x}}{\partial \vec{\xi}} &= \begin{pmatrix} -h & h \\ -h & 0 \end{pmatrix}, & \frac{\partial \vec{\xi}}{\partial \vec{x}} &= h^{-2} \begin{pmatrix} 0 & -h \\ h & -h \end{pmatrix} \\ G &= \begin{pmatrix} h^{-2} & -h^{-2} \\ -h^{-2} & 2h^{-2} \end{pmatrix}, & G : G &= 7h^{-4} \end{aligned} \quad (1.64)$$

473 It can be seen clearly, that  $G$  attains different values for the same physical triangle.  
474 This issue is best seen in simulations of low Re flows, where  $G : G$  is the dominant  
475 factor in the stabilization. The pressure field has a notable irregularity that however  
476 does not compromise stability. It has been observed, that this phenomenon is quickly  
477 masked, when the timesteps become smaller or when convection starts dominating.

#### 478 **Proposal: Principal Component Stabilization Tensor**

479 In face of the previously explained inconsistency, a new tensor is proposed, whose  
480 main purpose is to eliminate the dependency of the nodal enumeration of the pre-  
481 vious tensor  $G$ , while preserving the shape and orientation information that is ex-  
482 ploited by the stabilization. To develop the latter statement further, take for in-  
483 stance  $\vec{u}^T G \vec{u}$ . When evaluated, this term will yield a scalar quantity proportional  
484 to the square of the velocity projected over the meshsize in its direction.

485 Alternatively to the reference element, one can consider a reference equilateral tri-  
486 angle. This specific linear element can be seen as one that has a negligible affinity  
487 for any orientation of the flow. Therefore, it could be intuitively sought to model  $G$   
488 as a transformation encoding stretching and orientation of any element with respect  
489 to an isotropic unit triangle. A 2D isotropic element with meshsize  $h$  will generate  
490 a circumsphere of radius  $r = h/\sqrt{3}$ , regardless of its orientation (in 3D  $r = h\sqrt{6}/4$ ).

491 This circumsphere can be mapped with some stretching and rotation to an ellipsoid  
 492 encasing any triangle or tetrahedron (see for instance Fig. 1.10). If we assume  
 493 that any triangle sharing the same minimal ellipse has the same effect on the dis-  
 494 cretization, then we can instead use the mapping from a reference circumsphere to the ellipse. This bear the advantage, that the ellipsoid retains the stretching and

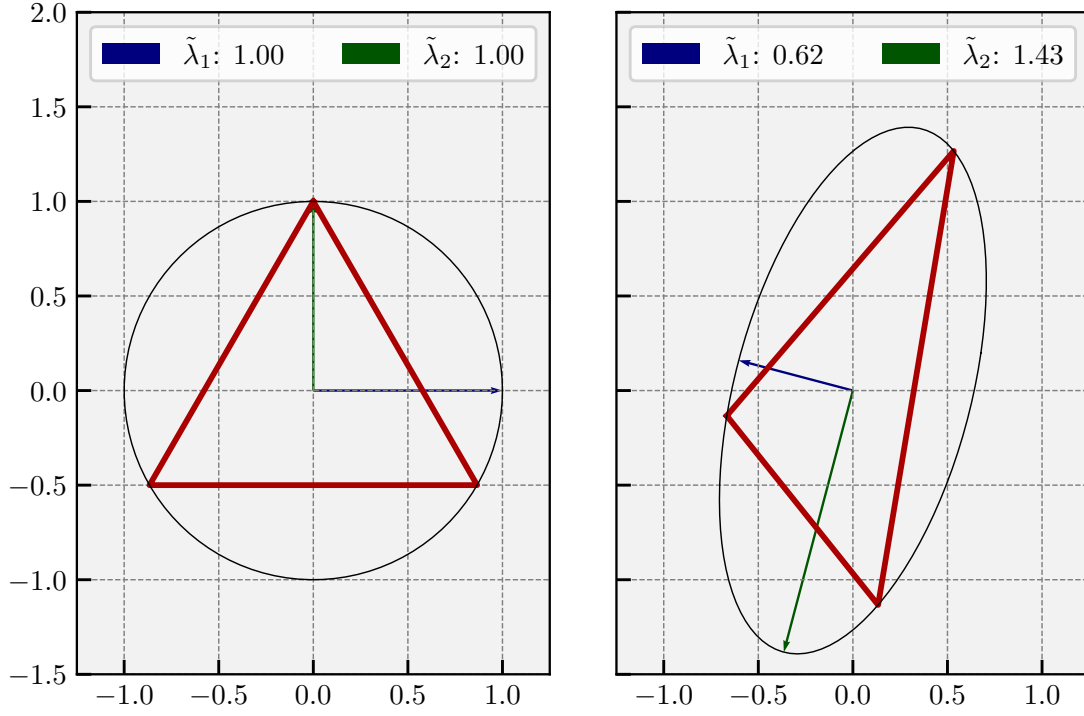


Figure 1.10: Two triangles in a 2D space (red) with their circumspheres (black) and principal axes (blue, green).

495 orientation information regardless of the order of the nodes spanning it. A mini-  
 496 mal ellipse, as shown in Fig. 1.9, can be readily computed with the eigenvalues and  
 497 eigenvectors of its nodes as followed. Take  $X \in \mathbb{R}^{n,d}$  as to be the non-square ma-  
 498 trix of nodal positions of any linear (or higher order) element in  $d > 1$  dimensions  
 499 centered around the origin. Then, its main directions (as exploited in the Principal  
 500 Component Analysis) are given by:  
 501

$$\vec{w}_1 = \operatorname{argmax}_{\|\vec{w}\|_2=1} \|X\vec{w}\|_2 = \operatorname{argmax}_{\vec{w}} \frac{\vec{w}^T X^T X \vec{w}}{\vec{w}^T \vec{w}} \quad (1.65)$$

502 In other words, if we look for the axis of largest variance of a set of points  $X$ ,  
 503 we need to solve the optimization given by Eq. 1.65. Luckily, we can avoid this  
 504 potentially expensive computation by considering for a short instance the Singular

505 Value Decomposition:

$$X^T X = (U \Sigma V^T)^T U \Sigma V^T = V^T \Sigma^T U^T U \Sigma V^T = V \Sigma^2 V^T \quad (1.66)$$

506 Verbally speaking, [Eq. 1.66](#) states that in order to find the principal components of  
507  $X$  we need to solve the eigenvalue problem of  $X^T X$ , which will yield the orthonormal  
508 column vectors  $w$  of  $V$  and the squared singular values in the quadratic matrix  $\Sigma^2$ .  
509 Note, that at the same time we confirm that  $X^T X$  yields an SPD matrix. With  
510 this knowledge, we can map the nodes on a circumcircle or radius 1 to the ellipse of  
511 another triangle with:

$$\begin{aligned} X_{ellipse}^{2D} &= X_{circle}^{2D} \sqrt{\frac{2}{3}} V \sqrt{\Sigma^2} V^T \\ X_{ellipsoid}^{3D} &= X_{sphere}^{3D} \sqrt{\frac{3}{4}} V \sqrt{\Sigma^2} V^T \end{aligned} \quad (1.67)$$

512 Another execution of the transformation can be seen in [Fig. 1.11](#), where the ref-  
513 erence equilateral tetrahedron on the left is transformed into the right one. This  
514 transformation maps from the reference frame to the physical frame, while  $G$  is  
515 supposed to be the square inverse mapping. Since our transformation is bijective,  
516 the inverse exists and is unique. Furthermore, it is avoided to compute a costly  
517 eigendecomposition for computational purposes.

$$G = \left( \frac{3}{2} 1_{\{d=2\}} + \frac{4}{3} 1_{\{d=3\}} \right) (X^T X)^{-1} \quad (1.68)$$

518 In the implementation of this method, the timestep component, which would in-  
519 fluence  $\tau_C$  through  $\tau_M$  is omitted, since in neither of the cases the derivation would  
520 yield such a term. In a latter subsection, this proposed model will be investigated  
521 in a reference case comparing it to more established methods, such as the one that  
522 follows.

### 523 Velocity projected meshsize

524 Another validated stabilization parameter is presented in works of Tezduyar *et*  
525 *al.* [\[26, 34\]](#). It bases on a velocity projected meshsize, much like before, but is  
526 instead computed using the gradient of the test functions  $\nabla \varphi_k$  of each node in an  
527 element:

$$h = \frac{\|\vec{u}_h\|}{\sum_{k \in K} |\vec{u}_h \cdot \nabla \varphi_k|} \quad (1.69)$$

528 The stabilization considers an element Reynolds number  $Re_{UGN}$  to switch between  
529 different stabilization coefficients in  $\tau_C$ . This term is also commonly referred to as

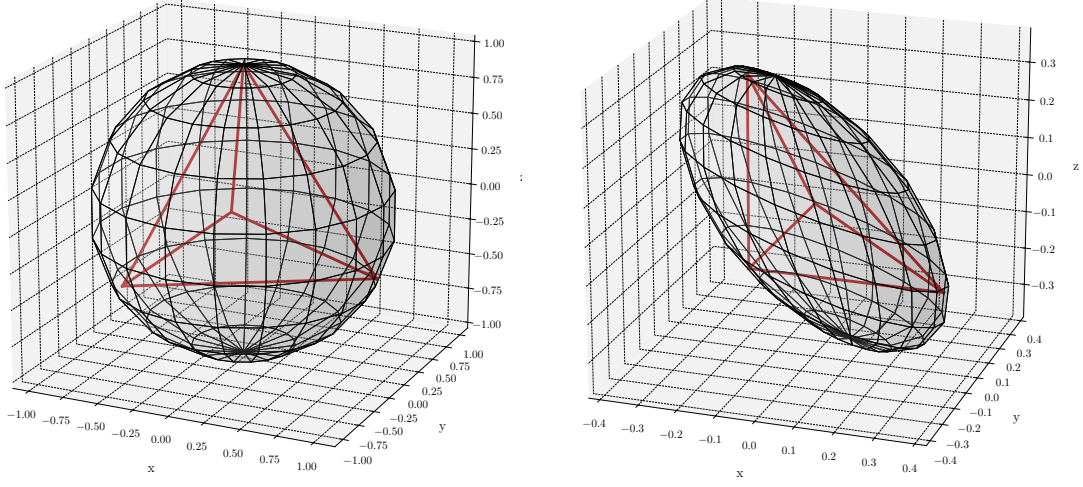


Figure 1.11: Execution of the transformation from a reference equilateral tetrahedron (left) to another one with the nodal coordinates  $(0,0,0)$ ,  $(0.5,0,0)$ ,  $(0,0.5,0)$ ,  $(0,0,0.5)$ . The corresponding stretching factors are  $\lambda_1 = 0.217$ , and  $\lambda_2 = \lambda_3 = 0.433$ .

the least squares on the incompressibility.

$$Re_{UGN} = \frac{\rho \|\vec{u}_h\| h}{\mu} \quad (1.70)$$

The resulting stabilization however requires a special treatment, if the velocities in the computation are or can be zero, as might be the case for instance at the beginning. There, it is recommended to use an average mesh size instead of the velocity projected mesh size (shown in Eq. 1.69).

$$\tau_M = \left[ \left( \frac{2\rho}{\Delta t} \right)^2 + \left( \frac{\rho \|\vec{u}_h\|}{h} \right)^2 + \left( \frac{\mu}{h^2} \right)^2 \right]^{-1/2} \quad (1.71)$$

$$\tau_C = \rho(1_{\{Re_{UGN} < 3\}} Re_{UGN}/3 + 1_{\{Re_{UGN} \leq 3\}}) \|\vec{u}_h\| h$$

### 1.3.4 Weak Formulation of the Large Scales

With the now cleared choice of subscales, they can be fed back into the large scale equations presented in Eq. 1.42. Before doing so, the equations will be rearranged using partial integration, to eliminate higher order derivatives and to simplify the

introduction of the subscales. Note, that as stated before, the subscales yield no boundary contributions by assumption.

$$\begin{aligned} \int_{\Omega} \vec{V}_h \cdot \mathcal{L}(\vec{U}_h + \vec{U}') dV &= \int_{\Omega} \vec{V}_h \cdot \vec{F} dV \\ \underbrace{\int_{\Omega} \vec{V}_h \cdot [\mathcal{L}(\vec{U}_h) - \vec{F}] dV}_{\text{Galerkin Terms}} &+ \underbrace{\int_{\Omega} \vec{V}_h \cdot \mathcal{L}(\vec{U}') dV}_{\text{Stabilization Terms}} = 0 \end{aligned} \quad (1.72)$$

Consider first the conventional Galerkin terms. The integrals are split using partial derivatives and transformed using the Gauß-Divergence Theorem to obtain boundary integrals. The boundary conditions are then applied, obtaining one surface integral that is active on boundaries with imposed stress conditions.

$$\begin{aligned} \int_{\Omega} \vec{V}_h \cdot [\mathcal{L}(\vec{U}_h) - \vec{F}] dV &= \int_{\Omega} \vec{w}_h \cdot \rho [\partial_t \vec{u}_h + (\vec{u}_h \cdot \nabla) \vec{u}_h - \vec{g}] dV + \\ &+ \int_{\Omega} \vec{w}_h \cdot [\nabla p_h - \nabla \cdot 2\mu \varepsilon(\vec{u}_h)] dV + \int_{\Omega} q \nabla \cdot \vec{u}_h dV \\ &= \int_{\Omega} \vec{w}_h \cdot \rho [\partial_t \vec{u}_h + (\vec{u}_h \cdot \nabla) \vec{u}_h - \vec{g}] dV + \\ &+ \int_{\Omega} -p_h \nabla \cdot \vec{w}_h + \mu \varepsilon(\vec{w}_h) : \varepsilon(\vec{u}_h) dV + \\ &+ \oint_{\partial\Omega} \vec{n} \cdot (pI - 2\mu \varepsilon(\vec{u}_h)) \cdot \vec{w}_h dA + \int_{\Omega} q \nabla \cdot \vec{u}_h dV \\ &= \int_{\Omega} \vec{w}_h \cdot \rho [\partial_t \vec{u}_h + (\vec{u}_h \cdot \nabla) \vec{u}_h - \vec{g}] dV + \\ &+ \int_{\Omega} -p_h \nabla \cdot \vec{w}_h + \mu \varepsilon(\vec{w}_h) : \varepsilon(\vec{u}_h) dV + \\ &+ \int_{\Omega} q \nabla \cdot \vec{u}_h dV + \int_{\Gamma_n} \vec{t}_n \cdot \vec{w}_h dA \end{aligned} \quad (1.73)$$

The stabilization terms are treated similarly except for the convective term, which is also transformed. This facilitates introducing the long expressions for  $\vec{U}'$  later.

$$\begin{aligned} \int_{\Omega} \vec{V}_h \cdot \mathcal{L}(\vec{U}') dV &= \int_{\Omega} \vec{w}_h \cdot \rho (\vec{u}_h \cdot \nabla) \vec{u}' + \vec{w}_h \cdot [\nabla p' - \nabla \cdot 2\mu \varepsilon(\vec{u}')] + q_h \nabla \cdot \vec{u}' dV \\ &= - \int_{\Omega} \vec{u}' \cdot [\rho (\vec{u}_h \cdot \nabla) \vec{w}_h + \nabla \cdot 2\mu \varepsilon(\vec{w}_h) + \nabla q_h] + p' \nabla \cdot \vec{w}_h dV \\ &:= \int_{\Omega} \vec{U}' \cdot \mathcal{L}(\vec{V}_h) dV \end{aligned} \quad (1.74)$$

The term  $\nabla \cdot 2\mu\varepsilon(\vec{w}_h)$  is commonly dropped at this point of the derivation assuming that “the tractions are continuous across inner element boundaries” [22]. Likewise, the diffusion term in the momentum residual is dropped due to the fact that it is 0 for linear elements. Introducing the subscales into Eq. 1.74 yields:

$$\begin{aligned} \int_{\Omega} \vec{V} \cdot \mathcal{L}(\vec{U}') dV &= - \int_{\Omega} \tau_M \vec{\mathcal{R}}_M \cdot [\rho(\vec{u}_h \cdot \nabla) \vec{w}_h + \nabla q_h] + \tau_C \mathcal{R}_C \nabla \cdot \vec{w}_h dV \\ &= \int_{\Omega} \tau_M [\rho \partial_t \vec{u}_h + \rho(\vec{u}_h \cdot \nabla) \vec{u}_h + \nabla p_h - \rho \vec{g}] \cdot \rho(\vec{u}_h \cdot \nabla) \vec{w}_h dV + \\ &\quad + \int_{\Omega} \tau_M [\rho \partial_t \vec{u}_h + \rho(\vec{u}_h \cdot \nabla) \vec{u}_h + \nabla p_h - \rho \vec{g}] \cdot \nabla q_h dV + \\ &\quad + \int_{\Omega} -\tau_C \nabla \cdot \vec{u}_h \nabla \cdot \vec{w}_h dV \end{aligned}$$

547 Since there is a regularity in the expressions above with the ones found in the  
 548 Galerkin terms, a common alias for the test functions acting on the convective term  
 549 is introduced  $\vec{w}_{supg} = \vec{w}_h + \tau_M \rho(\vec{u}_h \cdot \nabla) \vec{w}_h$ .

$$\begin{aligned} \int_{\Omega} \vec{V}_h \cdot \mathcal{L}(\vec{U}_h + \vec{U}') dV &= \int_{\Omega} \vec{w}_{supg} \cdot \rho [\partial_t \vec{u}_h + (\vec{u}_h \cdot \nabla) \vec{u}_h - \vec{g}] dV + \\ &\quad + \int_{\Omega} -p_h \nabla \cdot \vec{w}_h + \mu \varepsilon(\vec{w}_h) : \varepsilon(\vec{u}_h) dV + \\ &\quad + \int_{\Omega} q \nabla \cdot \vec{u}_h dV + \int_{\Gamma_n} \vec{t}_n \cdot \vec{w}_h dA \\ &\quad + \int_{\Omega} \tau_M \rho(\vec{u}_h \cdot \nabla) \vec{w}_h \cdot \nabla p_h dV + \\ &\quad + \int_{\Omega} \tau_M [\rho \partial_t \vec{u}_h + \rho(\vec{u}_h \cdot \nabla) \vec{u}_h + \nabla p_h - \rho \vec{g}] \cdot \nabla q_h dV + \\ &\quad + \int_{\Omega} -\tau_C \nabla \cdot \vec{u}_h \nabla \cdot \vec{w}_h dV \end{aligned} \tag{1.75}$$

550 The SUPG test function  $\vec{w}_{supg}$  has a didactic visual interpretation, as shown for a  
 551 simple 1D example in Fig. 1.12. The functions contain an upwind component, *i.e.* a  
 552 tilt in the weight towards the provenience of the velocity that scales with the gradient  
 553 in the velocity direction and stabilization parameter  $\tau_M$ . In other words,  $\vec{w}_{supg}$  acts  
 554 like an adaptive Upwind scheme that reacts to the flow conditions providing enough  
 555 diffusion in flow direction to maintain stability, while retaining accuracy where the  
 556 convection is minor. This method is likewise exploited in other element pairings,  
 557 such as  $P2$ - $P1$ , although no incompatibility between velocity and pressure shape  
 558 functions are present there. The pressure-velocity compatibility in the presented

559 VMS-type method comes from the term  $\tau_M \nabla p_h \cdot \nabla q_h$ . The reason why this term  
 560 adds control over the pressure can be best seen by considering the linear system  
 561 that follows from the discrete system. Take  $A$  as to be a matrix composed by the  
 562 blocks  $A_{\vec{u}, \vec{w}}$ ,  $A_{p, \vec{w}}$ ,  $A_{\vec{u}, q}$ ,  $A_{p, q}$ . The subscripts of the block denote the functions that  
 563 appear in it.

564

565  $A_{\vec{u}, \vec{w}}$  contains the discretization of:

$$\begin{aligned} & \int_{\Omega} \vec{w}_{supg} \cdot \rho [\partial_t \vec{u}_h + (\vec{u}_h \cdot \nabla) \vec{u}_h] - \tau_M \rho (\vec{u}_h \cdot \nabla) \vec{w}_h \cdot \rho \vec{g} dV + \\ & + \int_{\Omega} \mu \varepsilon(\vec{w}_h) : \varepsilon(\vec{u}_h) - \tau_C \nabla \cdot \vec{u}_h \nabla \cdot \vec{w}_h dV \end{aligned} \quad (1.76)$$

566  $A_{p, \vec{w}}$  contains the discretization of:

$$\int_{\Omega} -p_h \nabla \cdot \vec{w}_h + \int_{\Omega} \tau_M \rho (\vec{u}_h \cdot \nabla) \vec{w}_h \cdot \nabla p_h dV \quad (1.77)$$

567  $A_{\vec{u}, q}$  contains the discretization of:

$$\int_{\Omega} q \nabla \cdot \vec{u}_h + \tau_M [\rho \partial_t \vec{u}_h + \rho (\vec{u}_h \cdot \nabla) \vec{u}_h] \cdot \nabla q_h dV \quad (1.78)$$

568  $A_{p, q}$  contains the discretization of:

$$\int_{\Omega} \tau_M \nabla p_h \cdot \nabla q_h dV \quad (1.79)$$

569 To complete the terms consider  $B_{\vec{w}}$  and  $B_q$  as to the the right-hand side of the  
 570 momentum and continuity equations.  $B_{\vec{w}}$  contains the discretization of:

$$\int_{\Gamma_n} -\vec{t}_n \cdot \vec{w}_h dA + \int_{\Omega} \vec{w}_h \cdot \rho \vec{g} dV \quad (1.80)$$

571  $B_q$  contains the discretization of:

$$\int_{\Omega} \tau_M \rho \vec{g} \cdot \nabla q_h dV \quad (1.81)$$

572 In a plain Galerkin discretization the square block matrix  $A_{p, q}$  is indeed 0 every-  
 573 where, requiring  $A_{p, \vec{w}}$  to provide the needed information to compute  $p$ . This, as  
 574 discussed before, is not generally the case in same order Lagrange elements. The

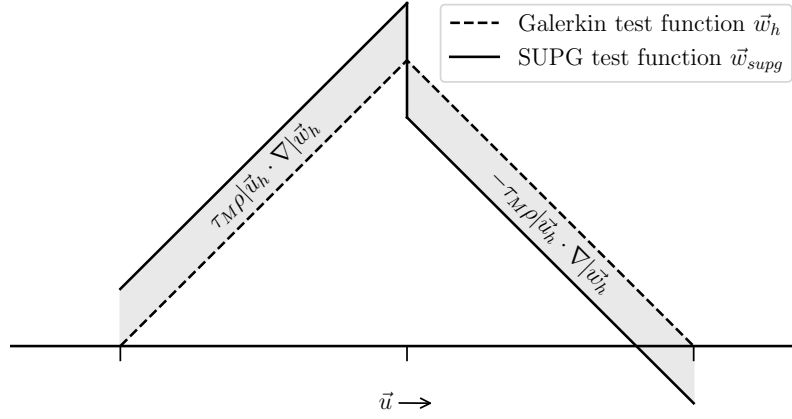


Figure 1.12: Scheme of a  $\vec{w}_{supg}$  test function for a 1D case. The contribution with respect to the Galerkin function  $\vec{w}_h$  is a piecewise-constant offset tilted towards the velocity's origin.

575 stabilization however adds non-zero contributions to  $A_{p,q}$  that make it SPD. The lat-  
 576 ter quality can be easily seen since the matrix is diagonally dominant with strictly  
 577 positive main diagonal entries. The last remaining component to discuss is the time  
 578 discretization of the problem. For this, a small demonstration will be done in the  
 579 following subsection considering different schemes for the time derivative, as well as  
 580 methods for treating the non-linearities of the problem.

### 581 1.3.5 Time Discretization

582 The time derivatives are commonly discretized using finite differences rather than  
 583 finite elements. The latter family of methods is also called Space-Time FEM and has  
 584 a superior complexity due to an added dimension in the finite element space. Finite  
 585 differences on the other hand are more straight forward, since the spatial structure  
 586 does not need to change drastically. The functions  $\vec{u}_h$  and  $p_h$  are approximated in our  
 587 case through a sum of scaled, piece-wise linear functions  $\varphi_i(\vec{x})$  satisfying  $\varphi_i(\vec{x}_j) = \delta_{ij}$   
 588 for any two nodes  $i, j$  in the mesh. The time dependency of the problem is shifted  
 589 onto the scaling factors, allowing for a separation of both problems.

$$\vec{u}_h = \sum_i \vec{u}_i(t) \varphi_i(\vec{x}), \quad p_h = \sum_i p_i(t) \varphi_i(\vec{x}) \quad (1.82)$$

590 For that reason, solving the problem can be seen as solving any system of the shape:

$$\partial_t u = f(t, u) \quad (1.83)$$

## Time Integration

A classic scheme used in hemodynamics simulations is the Backward Differentiation Formula (BDF). It is a per-definition implicit time scheme, *i.e.* it uses a right hand side that is defined at the unknown timestep  $t^{n+1}$  ( $t^n = n \Delta t$ ). The order of convergence of the scheme depends on the number of terms that are considered of  $u$  to build the derivative. For a simple first order BDF1 scheme, also simply called Backwards Euler method, develop a Taylor series around  $u^n$  considering the information known at  $u^{n+1}$ :

$$\begin{aligned} u^n &= u^{n+1} - \partial_t u^{n+1} \Delta t + \mathcal{O}(\Delta t^2) \\ \partial_t u^{n+1} &= \frac{u^{n+1} - u^n}{\Delta t} + \mathcal{O}(\Delta t) \end{aligned} \tag{1.84}$$

The obtained integration scheme is implicit and A-stable, or in simpler terms speaking, it enjoys a special robustness with respect to the time step size. Non-A-stable methods, as all explicit integration rules, is more tightly bound to smaller timesteps, since the integration can otherwise diverge. If another development is added with respect to two prior timesteps, the convergence order can be raised to two while maintaining A-stability. For that subtract 4 times Eq. 1.86 from Eq. 1.85.

$$u^{n-1} = u^{n+1} - 2\partial_t u^{n+1} \Delta t + 2\partial_{tt} u^{n+1} \Delta t^2 + \mathcal{O}(\Delta t^3) \tag{1.85}$$

$$u^n = u^{n+1} - \partial_t u^{n+1} \Delta t + \frac{1}{2} \partial_{tt} u^{n+1} \Delta t^2 + \mathcal{O}(\Delta t^3) \tag{1.86}$$

$$u^{n-1} - 4u^n = -3u^{n+1} + 2\partial_t u^{n+1} \Delta t + \mathcal{O}(\Delta t^3) \tag{1.87}$$

$$\partial_t u^{n+1} = \frac{3u^{n+1} - 4u^n + u^{n-1}}{2\Delta t} + \mathcal{O}(\Delta t^2) \tag{1.88}$$

BDF2 (Eq. 1.88) has some major benefits that motivate its choice in this work. First, its computational cost is minimal, involving only the storage of one additional history variable and few addition and multiplication operations. As a second order method, it maintains an overall small diffusive effect over the fluid dynamics. Lastly, it also benefits the solution of the linear system since it enforces the diagonal dominance on the  $A_{\vec{u}, \vec{w}}$  block over BDF1 for instance.

## Treatment of the non-linearities

Using implicit schemes benefits stability, but requires handling non-linear terms. In the stabilized Navier-Stokes equations there are several non-linearities involving shear-dependent viscosities, SUPG test functions and lastly, and most importantly, the non-linear velocity convection. The problem can be rewritten as a root-finding

619 problem for  $\vec{f}(\vec{u})$ , where  $A$  is the fully discretized system of equations and  $b$  the  
 620 right-hand side.

$$\vec{f}(\vec{u}) = A(\vec{u})\vec{u} - \vec{b} \quad (1.89)$$

621 The main approach followed when facing such a problem is finding an iterative rule  
 622 that provides improving guesses on the non-linear variable at the next timestep  $n+1$ .

623 **Picard iterations** (or Fixed-point iterations) are the most basic iterative ap-  
 624 proach. The system is solved using an initial guess  $\vec{u}^k = \vec{u}^n$  and without updating  
 625 the right hand side, the new solutions are plugged in as new  $\vec{u}^k$  in hopes to converge  
 626 towards an approximate  $\vec{u}^{n+1}$ .

$$\vec{u}^{k+1} = \vec{g}(\vec{u}^k) = A^{-1}(\vec{u}^k)\vec{b} \quad (1.90)$$

627 This methods bases on two conditions, first being that there exists a closed subset  
 628  $\mathcal{D}$  in which if  $\vec{u} \in \mathcal{D}$  then also  $\vec{f}(\vec{u}) \in \mathcal{D}$ . Secondly,  $\vec{f}(\vec{u})$  needs to be a contraction,  
 629 *i.e.* that

$$\|\vec{g}(\vec{x}) - \vec{g}(\vec{y})\| \leq \alpha \|\vec{x} - \vec{y}\| \quad (1.91)$$

630 for  $\alpha < 1$ ,  $x, y \in \mathcal{D}$ . In other words, that  $\vec{f}$  needs to be Lipschitz continuous with  
 631 a constant smaller 1 to ensure that iterating following Eq. 1.90 does not leave the  
 632 space containing the fixed-point. This can be shown by developing the iteration  
 633 rule. Suppose that  $\vec{u}$  is indeed a fixed-point of Eq. 1.90 and that all the conditions  
 634 above are met.

$$\begin{aligned} \vec{u} - \vec{u}^{k+1} &= \vec{g}(\vec{u}) - \vec{g}(\vec{u}^k) \\ \|\vec{u} - \vec{u}^{k+1}\| &\leq \|\vec{g}(\vec{u}) - \vec{g}(\vec{u}^k)\| \\ &\leq \alpha \|\vec{u} - \vec{u}^k\| \\ &\leq \alpha^k \|\vec{u} - \vec{u}^0\| \end{aligned} \quad (1.92)$$

635 The first order convergence in the sense of these methods is clearly only provided  
 636 for an  $\alpha < 1$ . The order of convergence is tight to the fact that  $\vec{g}(\vec{u}^k)$  is a first  
 637 order approximation of  $\vec{u}^{k+1}$ . This can be quickly seen by building a Taylor ex-  
 638 pansion around  $\vec{u}^{k+1}$  and neglecting any derivative terms. Picard iterations have  
 639 as an advantage that their implementation cost is minimal compared to other more  
 640 performing approaches, like the Newton-Raphson method. The latter, which can  
 641 also be seen as higher order fixed-point method, requires an expensive computation  
 642 of a Jacobian and its inverse. When handling systems, whose dynamics are highly  
 643 non-linear it is very important to implement such a routine to insure a proper reso-  
 644 lution of the underlying physics. In our simulations, we will be targeting flow with

notably low Reynolds numbers peaking approximately at  $Re = 1000$ . In this range, it is possible to use an alternative approach to Newton-Raphson, that is said to have a second order convergence while its implementation only adds a few terms to the linear system that can then be solved like in a fixed-point method.

**Linearized Newton** steps employ a second order development of the convective term as followed:

$$\begin{aligned}
 (\vec{u} \cdot \nabla) \vec{u} &= (\nabla \vec{u}) \cdot \vec{u} \\
 (\nabla \vec{u})^{k+1} \cdot \vec{u}^{k+1} &= (\nabla \vec{u})^k \cdot \vec{u}^k + ((\nabla \vec{u})^{k+1} - (\nabla \vec{u})^k) \cdot \vec{u}^k + \\
 &\quad + (\nabla \vec{u})^k \cdot (\vec{u}^{k+1} - \vec{u}^k) + \mathcal{O}(\|\vec{u}^{k+1} - \vec{u}^k\|^2) \\
 &= -(\nabla \vec{u})^k \cdot \vec{u}^k + (\nabla \vec{u})^{k+1} \cdot \vec{u}^k + (\nabla \vec{u})^k \cdot \vec{u}^{k+1} + \mathcal{O}(\|\vec{u}^{k+1} - \vec{u}^k\|^2)
 \end{aligned} \tag{1.93}$$

Where we assumed  $\vec{u}$  to have a Lipschitz-continuous gradient  $\nabla \vec{u}$  to merge the second order error terms into the present one. Under the conditions for regular convergence of fixed-point iterations and the latter condition, substituting the unknown  $(\nabla \vec{u})^{k+1} \cdot \vec{u}^{k+1}$  by the iteration rule Eq. 1.93 yields quadratic convergence. In practice it has been seen that the method is quadratic in convergence, however it has a smaller tolerance in regards to the initial error. Therefore, we have seen that too large time-steps or a cold initial start with a non-physical initial condition are indeed problematic for this iteration rule.

**Semi-implicit time integration** is an equally present choice in the field of computed hemodynamics. The principle is simple here, using a previous' timestep velocity in the integration.

$$(\vec{u}^{n+1} \cdot \nabla) \vec{u}^{n+1} = (\vec{u}^n \cdot \nabla) \vec{u}^{n+1} + \mathcal{O}(\Delta t) \tag{1.94}$$

The basic semi-implicit timesteps in theory introduce an error that scales linearly with the size of timesteps. In practice, it does however not greatly affect the convergence behaviour of the overall systems, even when the time derivatives are of higher order [35]. Nevertheless, it is possible to improve the scheme from a convergence-order point of view. For this purpose the linearization includes the information of a

previous timestep.

$$\begin{aligned} u^n &= u^{n+1} + \frac{\partial u(t^{n+1})}{\partial t} \Delta t + \mathcal{O}(\Delta t^2) \\ u^{n-1} &= u^{n+1} + \frac{\partial u(t^{n+1})}{\partial t} 2\Delta t + \mathcal{O}(\Delta t^2) \\ u^{n+1} &= 2u^n - u^{n-1} + \mathcal{O}(\Delta t^2) \end{aligned} \tag{1.95}$$

The Newton-Gregory backward polynomial (Eq. 1.95), if used, is usually matched by an equal order BDF scheme, since both require the same quantity of information and yield a proportional convergence order. When inspecting Eq. 1.95, it can be seen that the sought velocity is extrapolated, giving a double weight to the current velocity and balancing it out with the past. This carries a benefit to any integration in which the velocity evolves continuously in smooth steps, but can become fatal if the CFL number is too large.

**Non-linearities in viscosity and stabilization parameters** have so far not been the focus of the discussion, due to the prominence of the convective term. Both terms can be dealt with in a similar fashion, however, sub-increments for the viscosity model will be neglected in this work due to their overall slow dynamics compared to the core of the flow.

## 1.4 Validation Cases

In the previous section, a VMS-type method has been derived using common models from literature and a new proposal for stabilization parameters. Further, well-known time integration schemes and treatment options for the non-linearities have been shown and briefly discussed with regards to their stability, convergence and implementation. To actually display the theoretical thoughts, two test cases have been examined during the course of this work. The simulation framework bases on an in-house C++ library that provides mesh handling tools, a message parsing interface parallelization and a series of FEM solvers, including one for the incompressible Navier-Stokes equations [36, 37]. In the frame of this thesis, the latter was extended by the second order BDF scheme, the Newton-Gregory extrapolation, as well as the complete computation of the tensor-based stabilizations (Eq. 1.68, Eq. 1.56).

### 1.4.1 Flow Past a Cylinder in 2D

The first case corresponds to the well known flow past a cylinder in 2D. This test case has been thoroughly examined by the community and permits a good comparison and verification of the used methods. The case was set up following the indications

of Main *et al.* [38], who implemented the case to validate a new shifted boundary  
method. The domain is given in Fig. 1.13 including the sizes of the cylinder and the  
edges in unit scale. In the case, a uniform velocity is imposed at the left inlet, while

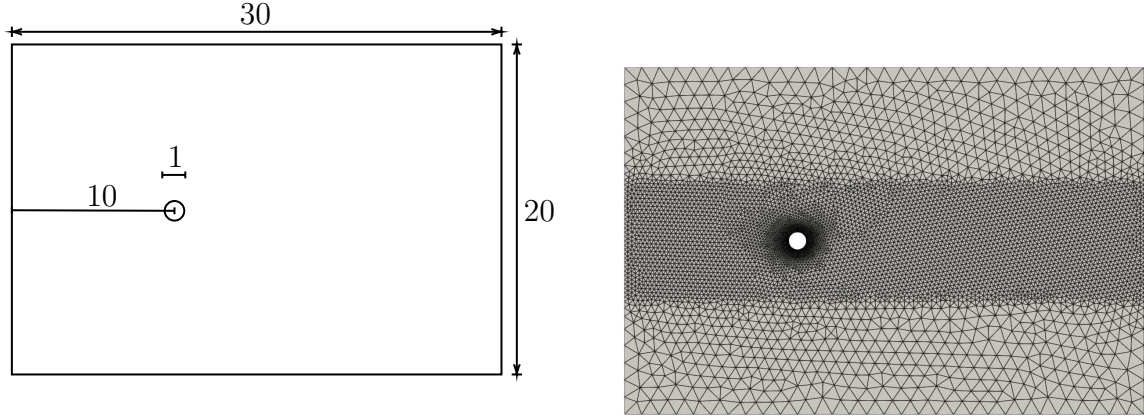


Figure 1.13: Case setup of the 2D flow past a cylinder. On the left is a sketch of the computational domain following the dimensions given in [38]. On the right is a view of the used mesh.

at the upper and lower boundaries enforce a zero normal velocity and no traction.  
The outlet on the right is set as a traction free boundary and the cylinder ( $d = 1$ )  
as a no-slip patch. The case is run for two well-studied Re numbers, 100 and 300.  
Both cause a regular formation of vortices in the wake of the cylinder, however  
for the latter Re, these are more developed (see Fig. 1.14 for better image of the  
phenomenon at  $Re = 300$ ). The simulations have been run for a unit Newtonian  
viscosity  $\mu$  and a unit density  $\rho$ . The control over the Re was thus exerted through  
the inflow velocity  $u_\infty$ .

$$Re = \frac{\rho u_\infty d}{\mu} \quad (1.96)$$

The generated mesh consists of 13622 triangle elements with a maximal isotropic  
edge length of  $h = 0.7$  in the upper and lower tranches and  $h = 0.25$  in the core of the  
domain. In a ring of thickness 1 around the cylinder the mesh is constructed with  
20 layers of structured elements, which blend into the non-structured surroundings  
with a linear mesh grading from  $h = 0.075$  to  $h = 0.25$ . The timestep size for the  
simulations was chosen to obtain approximately a maximal CFL number of 1.

$$CFL = \|\vec{u}\| \frac{\Delta t}{h} \quad (1.97)$$

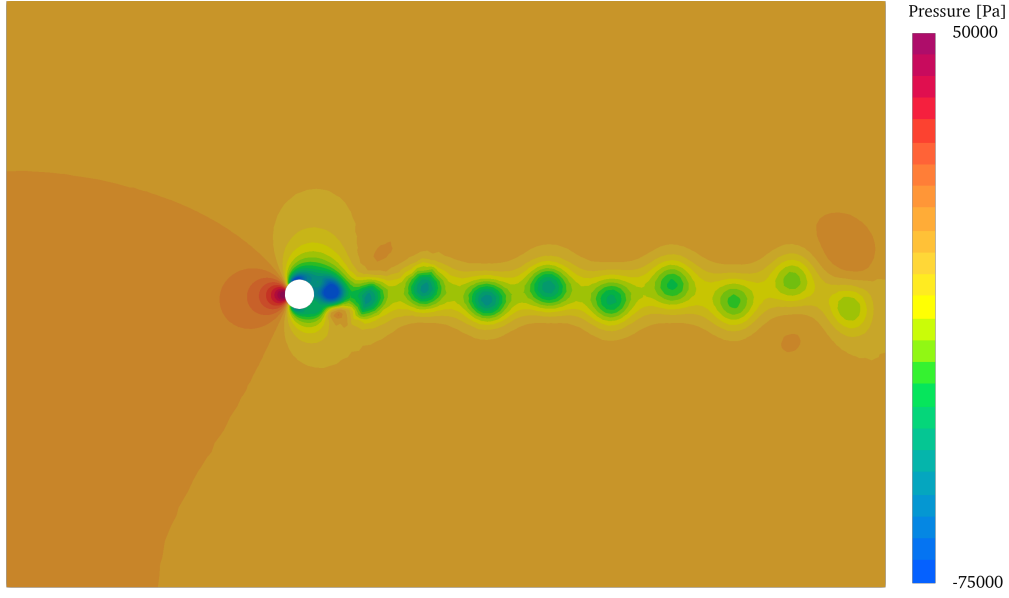


Figure 1.14: Pressure field taken during the simulation at  $Re = 300$ .

713 **The time integration** was first inspected, running simulations with the first and  
714 second order BDF schemes on the case with  $Re = 300$ . To isolate the effect of the time  
715 scheme, the timestep was frozen in the stabilization parameters to the minimal value  
716 of  $10^{-5}$ . The simulations were run for  $CFL = \{1, 2, 4, 8\}$  and the phase shift between  
717 the formation of the first vortices in each case was computed. In the simulations  
718 there are two main phenomena that can be seen when varying the schemes and  
719 timesteps. Firstly, more diffusive schemes and larger timesteps, consistently push  
720 the generation of the first vortex to later stages of the simulation. While both  
721 schemes show their expected order of convergence, as can be seen in Fig. 1.15, the  
722 second order scheme shows overall much smaller delays. This motivates the choice  
723 of BDF2 for the rest of the tests, since no measurable increase in computation time  
724 could be recorded and all the simulations showed a stable behaviour.

725 **Drag and lift** comparisons are the next important element to consider. The drag  
726 number, or drag coefficient ( $c_D$ ), is a dimensionless quantity that represents the drag  
727 force acting on an object moving through a fluid.

$$c_D = \frac{2F_D}{\rho u_\infty^2 d} \quad (1.98)$$

728 where  $F_D$  is the drag force exerted by the fluid onto the cylinder surface. The lift  
729 number, or lift coefficient ( $c_L$ ), is used to quantify the lift generated by an object,  
730 such as an airfoil, in a fluid flow. It depends on same quantities as the drag, excepting

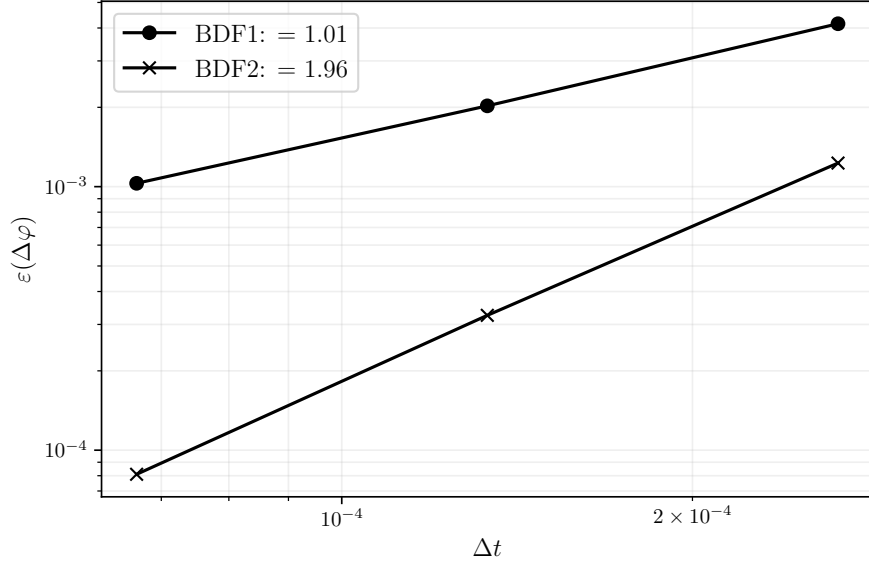


Figure 1.15: Log-log plot of the phase shift error  $\varepsilon(\Delta\varphi)$  against the timestep size.

the force, which is in this case the one acting in vertical direction ( $F_L$ ). Naturally, the lift of the flow past a cylinder should be on average 0. Even though it is arguably an objective evaluation criterion, it is seldomly included in studies. Nevertheless, it is provided below for a better reference.

$$c_L = \frac{2F_L}{\rho u_\infty^2 d} \quad (1.99)$$

Lastly, the Strouhal number (St) is considered, which is likewise a dimensionless number that describes oscillating flow mechanisms. It is used to characterize the frequency of vortex shedding and other periodic flow phenomena as the present one.

$$St = \frac{fd}{u_\infty} \quad (1.100)$$

where  $f$  is the frequency of vortex shedding that in our case has been computed from a Fast-Fourier-Transformation of  $F_L$ .

The corresponding plots displaying both lift and drag for both studied Re are seen in Fig. 1.16. For  $Re = 300$ , both stabilizations yield a notoriously similar behaviour, suggesting that for larger Re numbers both techniques have a very similar effect. The average drag force, recorded in Table 1.2, is slightly closer to reference values in the newly proposed stabilization, while the St number is larger in the third digit. For  $Re = 100$ , the oscillatory profiles of both stabilizations differ both in phase and

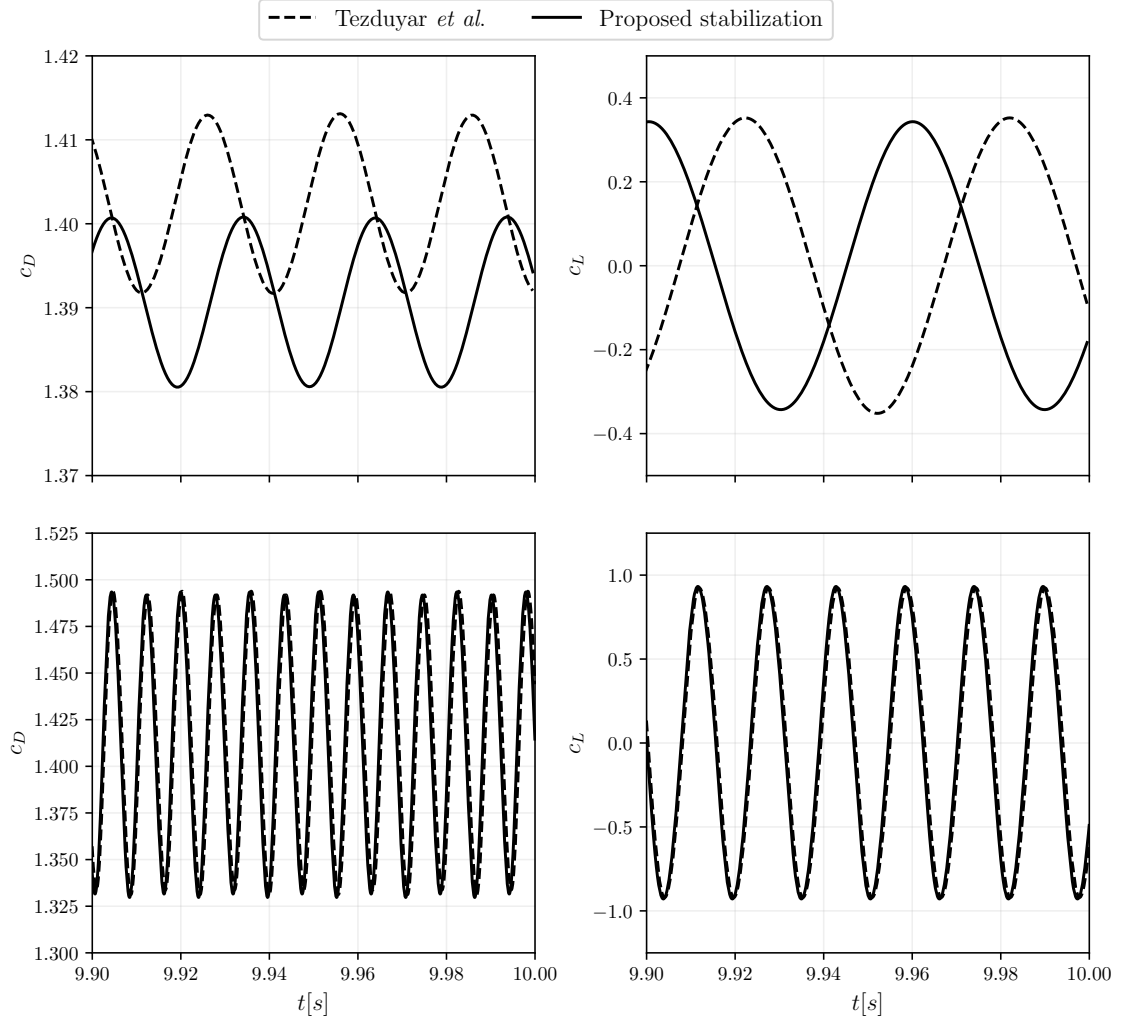


Figure 1.16: Drag (left) and lift (right) computed on the cylinder at  $Re = 100$  (top) and  $Re = 300$  (bottom) for the computations using the preimplemented stabilization (Tezduyar et al. [26]) and the newly proposed principal component tensor stabilization.

| Re = 100 | Proposed stab. | Tezduyar <i>et al.</i> | Ref. [26]     | Ref. [39] | Ref. [38] |
|----------|----------------|------------------------|---------------|-----------|-----------|
| $c_D$    | 1.391          | 1.402                  | 1.417 - 1.422 | 1.335     | 1.35      |
| $c_L$    | 2e-4           | 1e-3                   | -             | -         | -         |
| St       | 0.168          | 0.167                  | 0.167 - 0.169 | 0.157     | 0.167     |
| Re = 300 | Proposed stab. | Tezduyar <i>et al.</i> | Ref. [39]     | Ref. [40] | Ref. [38] |
| $c_D$    | 1.414          | 1.415                  | 1.367         | 1.38      | 1.38      |
| $c_L$    | 1e-3           | 8e-4                   | -             | -         | -         |
| St       | 0.214          | 0.213                  | 0.215         | 0.213     | 0.211     |

Table 1.2: Table summarizing the computed drag, lift and Strouhal numbers for  $Re = 100$  and  $Re = 300$ . The two first columns contain values obtained from the simulations using the principal component metric tensor (Eq. 1.68) and the previously implemented stabilization from [26]. On the right are reference values from literature.

mean value. The same trends from before can be observed, however with larger absolute differences. Here, the newly introduced stabilization tends overall more towards literature values excepting reference [26], which implements the same method as in the second column. The St number is, following [26], ideally large for less diffusive simulations, also supporting the new contribution. This leads to the following remarks:

### Conclusions

- Considering the coarse computation of the case compared to the complex reference computations (high order elements, increased mesh refinement, etc), both implemented methods yield satisfactory results.
- The proposed stabilization passes a contribution to the continuity stabilization that is dominant in the low Re regime, which can be also observed in the simulations. The better match with literature reassures the choice.
- Second order time integration bears larger benefits at negligible costs, thus making it the default choice unless stated otherwise.

752

## 1.4.2 FDA Nozzle

The FDA Nozzle, proposed by the U.S. Food and Drug Administration (FDA), is a simplified, idealized medical device geometry used for benchmarking CFD codes. It shares characteristics with various blood-carrying medical devices, such as catheters and syringes. The nozzle includes a radial step, sharp edges, and a cross-sectional

stenosis to induce shear stresses and simulate conditions like hemolysis [41]. The axisymmetric design features a 40 mm long stenotic throat with a 20° connecting cone at one end and a step change in diameter at the other (see Fig. 1.17 for more details). This configuration creates regions of accelerating and decelerating flow, shear stress

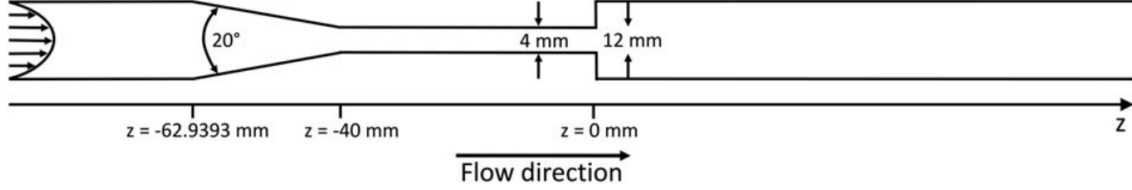


Figure 1.17: Design of the FDA Nozzle. Taken from [42].

variations, and recirculating flow, mimicking the flow dynamics observed in medical devices. Although medical devices are not part of the context of this thesis, this case presents interesting CFD challenges to tackle. The numerous studies handling it provide a didactic explanation on how different approaches, such as turbulence models, influence the obtained results within different Re regions. For the analysis, throat Reynolds numbers of  $Re = 500$  and  $Re = 2000$  will be considered. The prior lays in a similar range as the bloodflow in brain arteries and showcasing a fully laminar flow. For that reason it is easier to validate the computational framework, since transient turbulent effects are not present. The case of  $Re = 2000$  on the other hand sheds light into the solvers behaviour in the transition zone. Here, in some cases the jet breaks down in the rear end of the nozzle causing turbulent effects. The appearance of this breakdown depends on the used turbulence models, mesh refinement and experimental setup, and is thus introduces an interesting discussion about to what degree it is reasonable or not to refine the computations.

## Case setup

The FDA Nozzle has been experimentally tested by 3 teams participating in the computational challenge by Stewart *et al.* [43]. The used liquid was a blood-like, Newtonian solution containing water, glycerin, and sodium iodide with a density of  $1056 \text{ kg/m}^3$ . The viscosity of the fluid is reported to be  $3.5 \text{ mPa}\cdot\text{s}$ , which resembles the plateau viscosity of blood used in many rheology models. In the numerical scenario, a constant parabolic inflow profile is applied, whose maximum velocity is set to match the following volumetric flow:

$$\dot{Q}_{Re500} = 5.21 \cdot 10^3 \text{ mm}^3/\text{s}, \quad \dot{Q}_{Re2000} = 2.08 \cdot 10^4 \text{ mm}^3/\text{s} \quad (1.101)$$

The walls of the nozzle are set to no-slip conditions and the outlet is rendered traction-free. The nozzle has been generated for an axial extension of 168 mm ranging from  $z = -88 \text{ mm}$  to  $z = 80 \text{ mm}$ . The interior of the nozzle was meshed with

unstructured tetrahaedra with a varying anisotropy in the axial direction, resulting in 3.8M elements. In the first three sections of the nozzle, elements are generated with an edglength of 0.2mm and an aspect ratio of 4 in the axial direction. In the converging section, the elements are naturally aligned with the narrowing walls. After the expansion, the aspect ratio is dropped to 2 due to the possibly more chaotic flow. The time discretization is done using BDF2 with regular timesteps at  $\Delta t = 0.5$  ms and  $\Delta t = 0.1$  ms for  $Re = 500$  and  $Re = 2000$ , respectively, to obtain approximately unit CFL numbers. Linearized Newton iterations are run between each timestep until the  $L_2$  initial residual norm drops below  $10^{-6}$ . The residual norm of the linear solver was set to the same value. The subsequent simulations were carried out using the newly proposed stabilization (Eq. 1.68).

## Results

Per-timestep convergence was obtained in the case of  $Re = 500$  within 4 iterations, while for  $Re = 2000$  the convergence depended on the stage of the simulations. Before attaining the steady-state profile shown in the lower half of Fig. 1.18, the simulation passed two phases. The first is given by the progressive advancement of the jet in the nozzle (as seen on the upper half of Fig. 1.18). Convergence speed there is above 2, while only 5 iterations are required to meet the set tolerance (as seen in Fig. 1.19). Once the jet reaches the outlet, it causes reentering flow through its

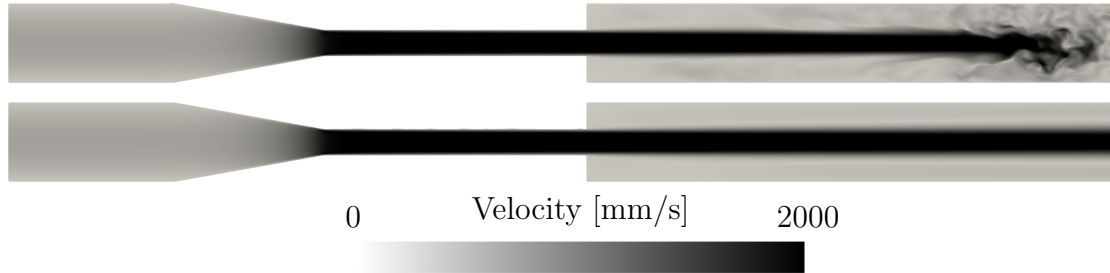


Figure 1.18: Opacity-coded velocity at  $Re = 2000$  at two states. Top: Initial transient phase with presence of vortex structures. Bottom: Converged stage past on which the results have been computed.

shedding vortices. The reflux in turn affects the local coercivity causing convergence issues. Once the vortices are evacuated, the flow starts approaching its steady-state profile. In the last section on the right of Fig. 1.19, the seemingly two parallel curves follow from an alternating 2 and 3 subiterations needed for convergence.

Once a steady-state was reached with an initial residual below  $10^{-6}$  for both  $Re$ , the results were extracted along the main axis of the nozzle, recording the axial velocity

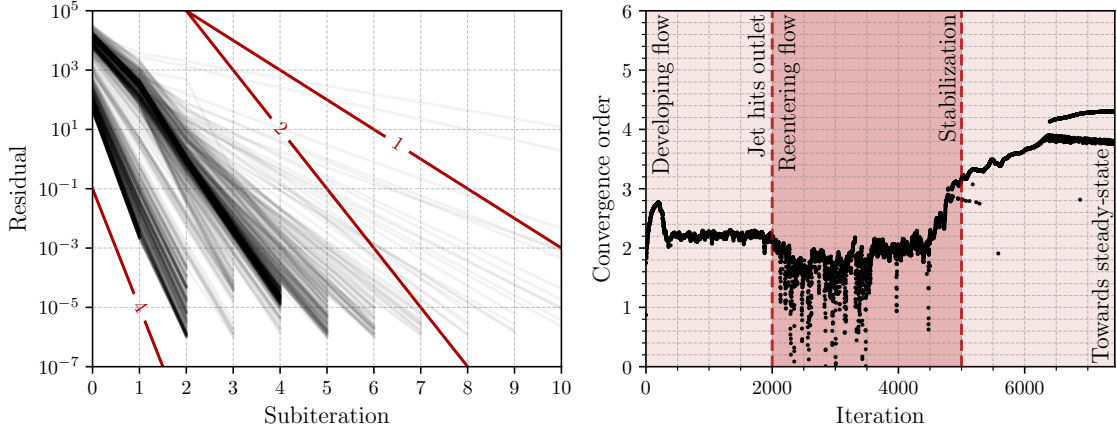


Figure 1.19: Non-linear convergence of the FDA Nozzle at  $Re = 2000$ . Left: Initial residual at the beginning of each subiteration. The last residual is not plotted and is below  $1e-6$ . One of every ten curves is shown with a fixed transparency. Right: Convergence order corresponding to the curves for all timesteps until a nearly steady-state is reached. The three sections mark (i) the travelling of the jet through the nozzle, (ii) the jet hitting the outflow boundaries, causing reflux and hindering convergence, and lastly the convergence towards the steady-state solution for which we report results.

812  $u_z$ , pressure and massflow. The latter was transformed into a percentile error using  
813 the theoretically imposed value as suggested in [43]:

$$E_Q = 100\% \times \frac{\dot{Q} - \dot{Q}_{theory}}{\dot{Q}_{theory}} \quad (1.102)$$

814 The obtained profiles for  $Re = 500$  in Fig. 1.20 show mass conservation errors be-  
815 low 0.6%. The pressures have been plotted including measurement data [43] and a  
816 reference computation. It can be clearly seen that there is a systematic mismatch  
817 between CFD computations for the pressure and experimental recordings. This miss-  
818 match however exists between the measurements and all reported simulations of the  
819 reference study [43]. This has been at least partly attributed to normalization errors  
820 in the experimental setup. The axial velocity at the centerline on the other hand  
821 fit the measurements consistently throughout the nozzle. Further, cross sections of  
822 the flow were compared to Particle Imaging Velocimetry (PIV) measurements at six  
823 locations:

$$z \in \{-88 \text{ mm}, -48 \text{ mm}, -8 \text{ mm}, 8 \text{ mm}, 24 \text{ mm}, 80 \text{ mm}\}$$

824 The resulting profiles were extracted along the positive x-axis values and shown in  
825 Fig. 1.21. The axial velocity is well reconstructed in the first four locations, but

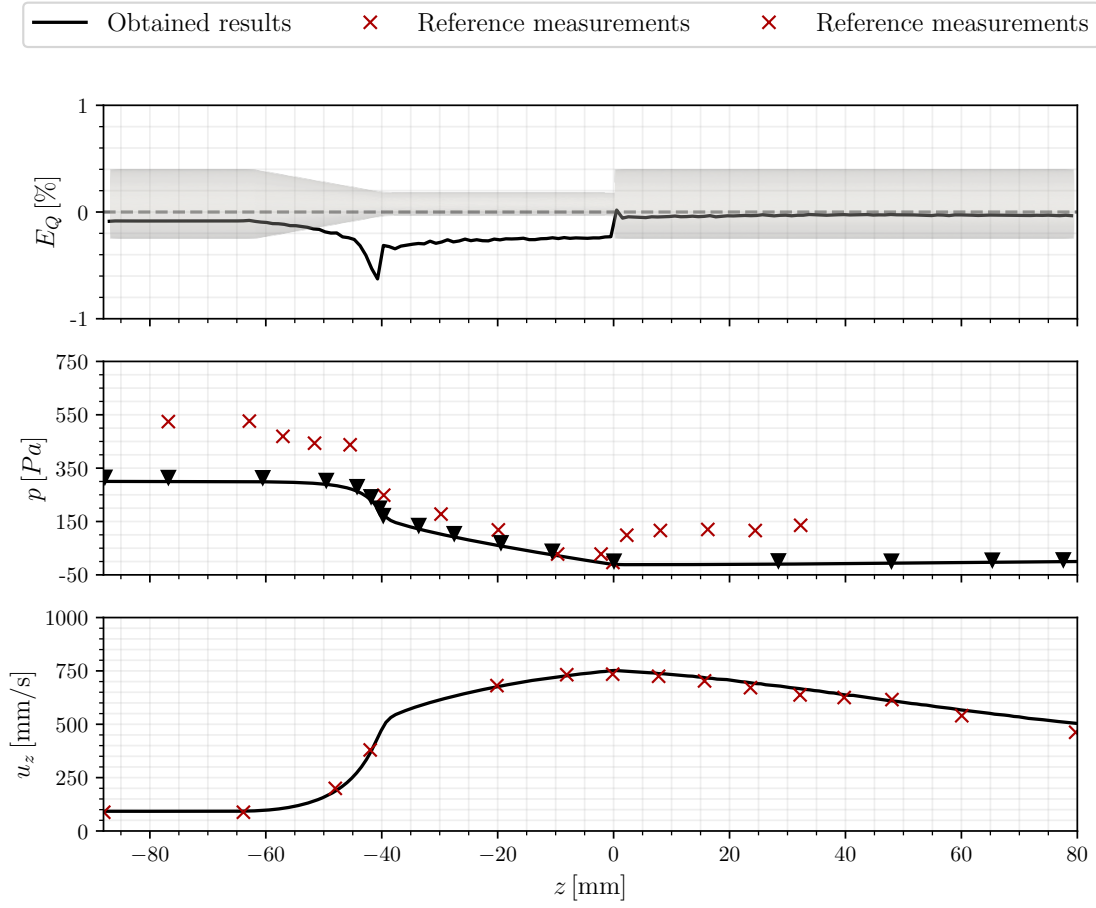


Figure 1.20: Resulting mass conservation errors, pressures and axial velocities along the main nozzle axis at  $Re = 500$ . Results are plotted against measurements and reference computations found in Stewart et al. [43].

shows a mismatch of approximately 50 mm/s at last recorded location. There, the measured values indicate a more strongly diffused pipeflow compared to the simulation. When considering the  $Re = 2000$ , the massflow is similarly well fit with a

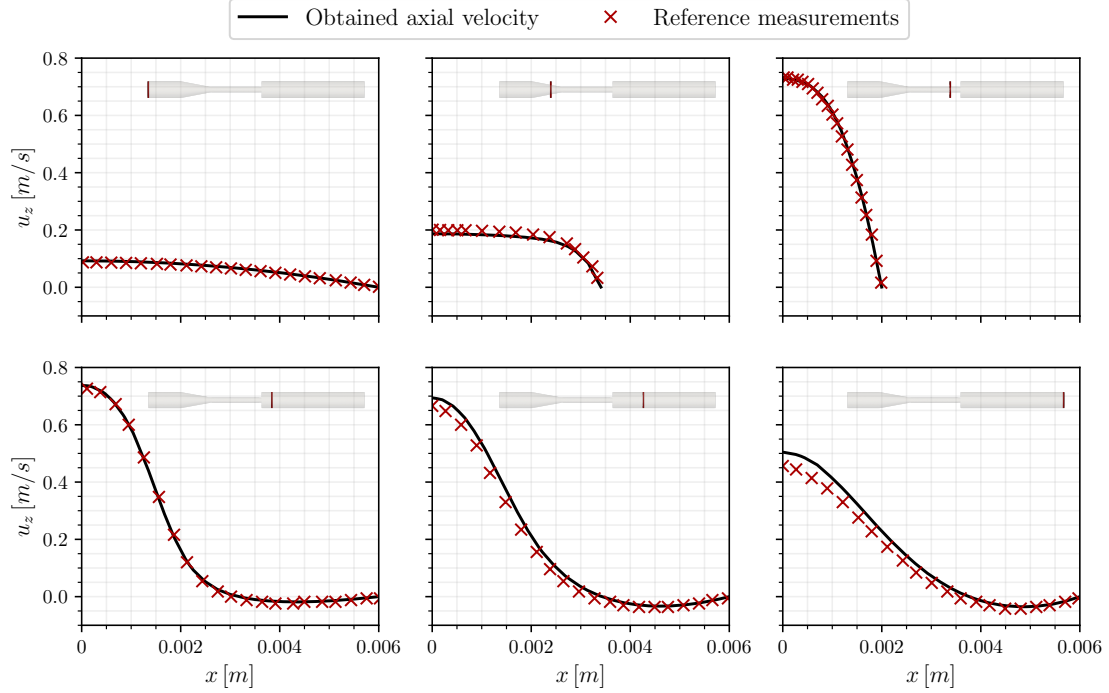


Figure 1.21: Axial velocity profiles of the FDA Nozzle at  $Re = 500$  on locations marked in red along the blended-in nozzle.

mean error below 0.2%. In this case, the pressure is well predicted by the simulations throughout the given values. The centerline velocity is like-wise well fit, excepting the sections past  $z = 48$  mm. There, the PIV returns a strongly decaying peak velocity that is subject to large variations among three different experimental setups. This suggests the breakdown of the jet and subsequent turbulent structures that diffuse kinetic energy in the stream. For reference, the results of a laminar simulation conducted for the previously mentioned study [43] are shown. There, as in our case (see Fig. 1.18), the jet does not yet break down retaining most of its structured properties from the previous sections. The breakdown can be also seen in the radial cross-sections, which this time are taken at slightly modified locations (again, based on the curves provided in [43]):

$$z \in \{-88 \text{ mm}, -80 \text{ mm}, 8 \text{ mm}, 32 \text{ mm}, 60 \text{ mm}, 80 \text{ mm}\}$$

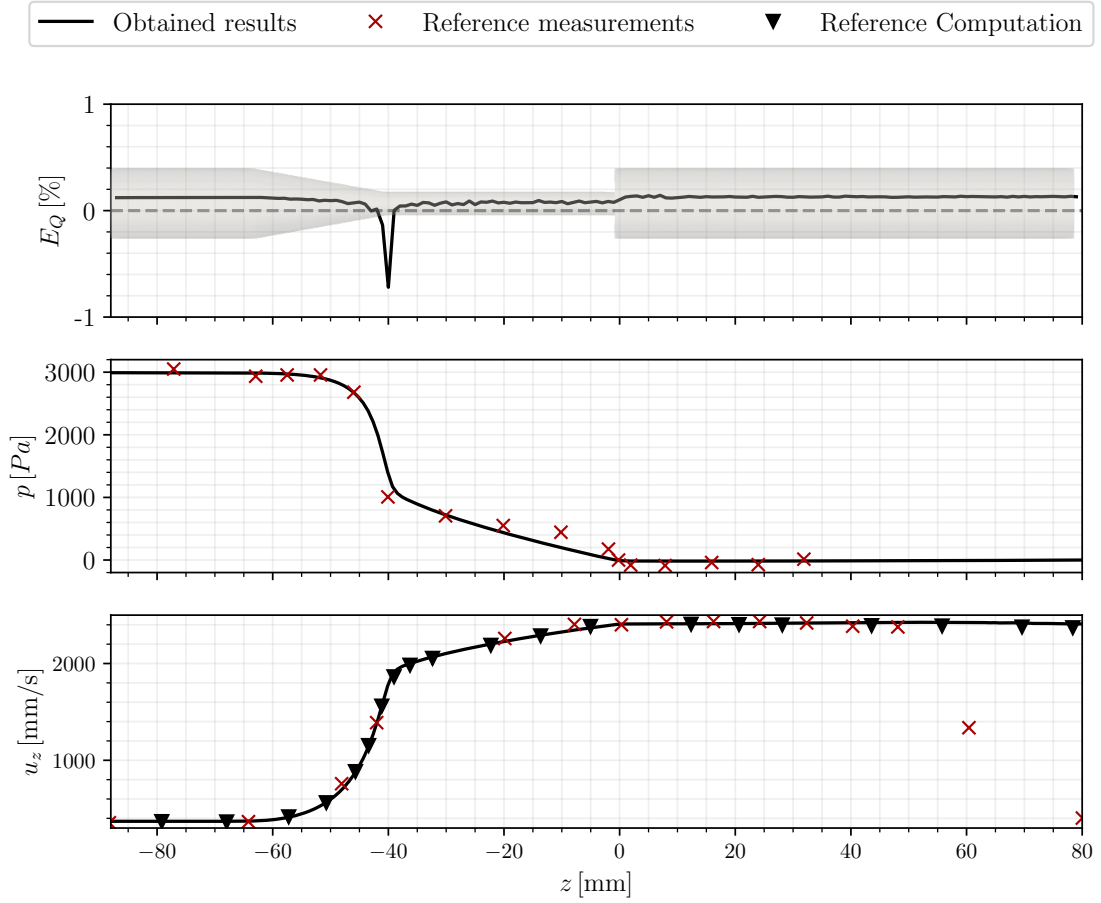
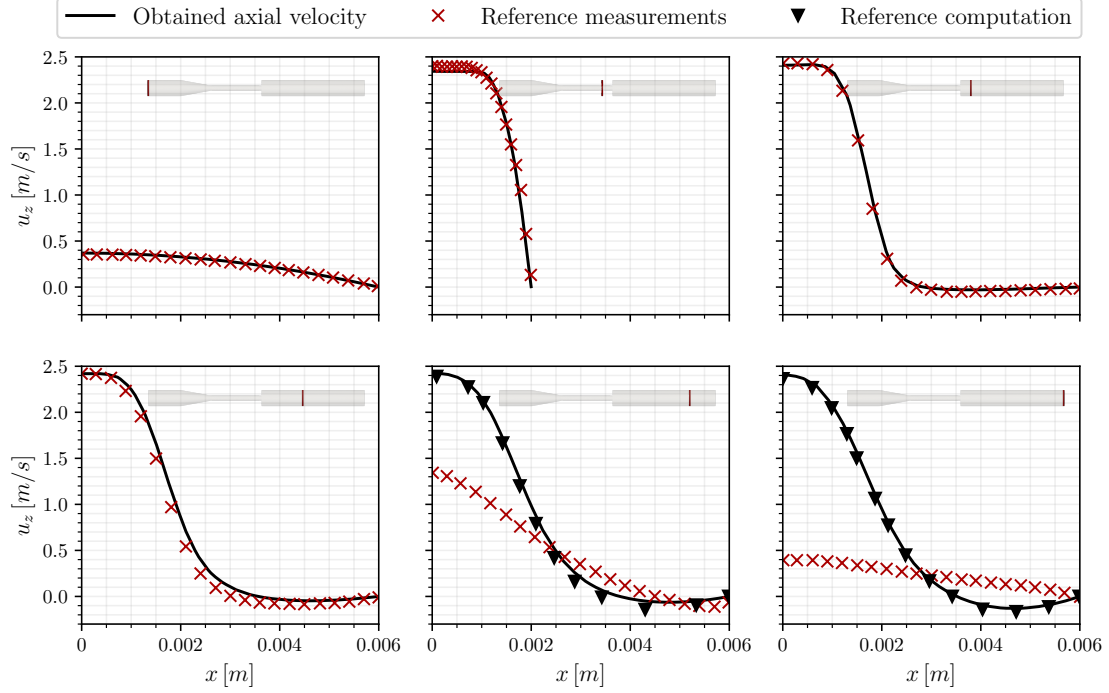


Figure 1.22: Resulting mass conservation errors, pressures and axial velocities along the main Nozzle axis at  $Re = 2000$ . Results are plotted against measurements and reference computations found in Stewart et al. [43].

Similarly to before, the first four sections confirm a good match between theory and practice. In the latter two sections, the jet flattens out in the experiment, while the conducted simulations of this work and the reference publication remain similar.



[Axial velocity profiles of the FDA Nozzle at  $Re = 2000$ ] Axial velocity profiles of the FDA Nozzle at  $Re = 2000$  on locations marked in red along the blended-in nozzle.

## Discussion

As stated before, this case serves two purposes. First, validating the numerical framework for a more simple, fully laminar setup at  $Re = 500$ , and secondly providing a perspective on the performance in a more complex flow regime,  $Re = 2000$ . The fully laminar matches experimental data and other reported results from laminar simulations [43], reassuring us about the validity of our method. Hereby, it is worth noting that the obtained results base on a mesh that is far from the resolutions that are otherwise used for benchmarking the FDA Nozzle [42, 44]. This is important to consider since simulations of hemodynamics cannot afford excessive computational demands.

For Reynolds numbers of 2000 and above there has been a large effort from turbulence research to better understand the mismatch among cutting edge CFD simulations and experiments, such as the ones referenced to before. In experiments, small

856 impurities in the initial flow or minor asymmetries in the mould can cause a diminute  
 857 perturbation of the flow along its path. This perturbation is naturally amplified by  
 858 the sudden expansion, causing the jet to degenerate into a turbulent structure. In  
 859 computations, asymmetries inherent to the mesh, the numerical method or simply  
 860 round-off errors are the ones triggering the same effect. The question is, at what  
 861 stage does this happen. Two highly recommendable studies [42, 44] employed differ-  
 862 ent numerical techniques to investigate this issue tracking the jet breakdown along  
 863 the nozzle. In both cases, it was concluded that the breakdown location is shifted  
 864 farther and farther towards the outlet when increasing the fidelity of the numerics  
 865 through mesh refinement and higher order schemes. This is a natural consequence  
 866 following from a higher spectrum of flow structures being simulated with increas-  
 867 ing accuracy. Bergersen *et al.* [42] expanded this analysis further by considering  
 868 upstream perturbations and their magnified effects downstream. This sketched a  
 869 reason why these experimental phenomena happen and why the disagreement in  
 870 appears in the transitional regime. Interested readers are incited to examine the  
 871 referenced publications for more details on this thrilling subject.

### Conclusions

*In this chapter, three main topics have been described and discussed: (i) the derivation of the equations dominating the flow of blood in human arteries, (ii) the numerical frameworks to approximate them and lastly (iii) two illustrative validation cases on which newly proposed methods were examined. With the numerical experiments yielded satisfactory results giving credit to the introduced tensor metric, but more extensive verification and convergence tests should be carried out to confirm its validity. Further, the experiments have served a didactic purpose during the course of the project, since thanks to them some optimal schemes and modelling options were identified, saving some time in setting up the framework for patient-specific simulations. These introduce namely other, but equally complex challenges to tackle.*

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## RÉSUMÉ

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Sed tamen haec cum ita tutius observentur, quidam vigore artuum inminuto rogati ad nuptias ubi aurum dextris manibus cavatis offertur, inpigre vel usque Spoletium pergunt. haec nobilium sunt instituta.

## MOTS CLÉS

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Intracranial aneurysms, Haemodynamics, Computational Fluid Dynamics, Finite Elements, Variational Multi-scale Method

## ABSTRACT

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Ego vero sic intellego, Patres conscripti, nos hoc tempore in provinciis decernendis perpetuae pacis habere oportere rationem. Nam quis hoc non sentit omnia alia esse nobis vacua ab omni periculo atque etiam suspicione belli?

## KEYWORDS

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Intracranial aneurysms, Haemodynamics, Computational Fluid Dynamics, Finite Elements, Variational Multi-scale Method